

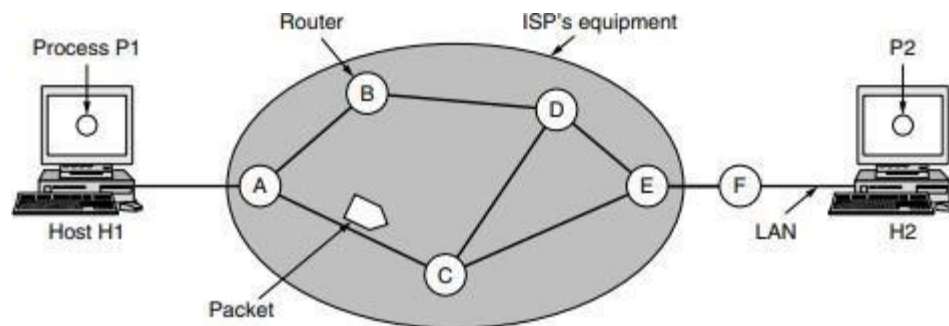
UNIT – III

NETWORK LAYER

Network Layer Design Issues:

1. Store-and-Forward Packet switching

Before starting to explain the details of the network layer, it is worth restating the context in which the network layer protocols operate. This context can be seen in Fig. 5-1. The major components of the network are the ISP's equipment (routers connected by transmission lines), shown inside the shaded oval, and the customers' equipment, shown outside the oval. Host H1 is directly connected to one of the ISP's routers, A, perhaps as a home computer that is plugged into a DSL modem. In contrast, H2 is on a LAN, which might be an office Ethernet, with a router, F, owned and operated by the customer. This router has a leased line to the ISP's equipment. We have shown F as being outside the oval because it does not belong to the ISP. For the purposes of this chapter, however, routers on customer premises are considered part of the ISP network because they run the same algorithms as the ISP's routers (and our main concern here is algorithms).



This equipment is used as follows. A host with a packet to send transmits it to the nearest router, either on its own LAN or over a point-to-point link to the ISP. The packet is stored there until it has fully arrived and the link has finished its processing by verifying the checksum. Then it is forwarded to the next router along the path until it reaches the destination host, where it is delivered.

2. Services provided to transport layer:

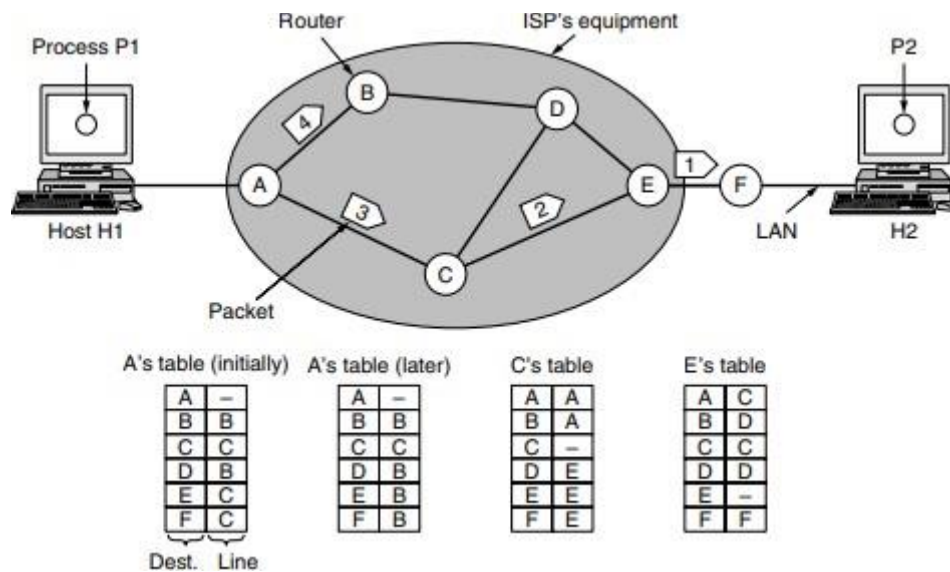
The network layer provides services to the transport layer at the network layer/transport layer interface. An important question is precisely what kind of services the network layer provides to the transport layer. The services need to be carefully designed with the following goals in mind:

1. The services should be independent of the router technology.
2. The transport layer should be shielded from the number, type, and topology of the routers present.
3. The network addresses made available to the transport layer should use a uniform numbering plan, even across LANs and WANs.

3. Implementation of connection-less service

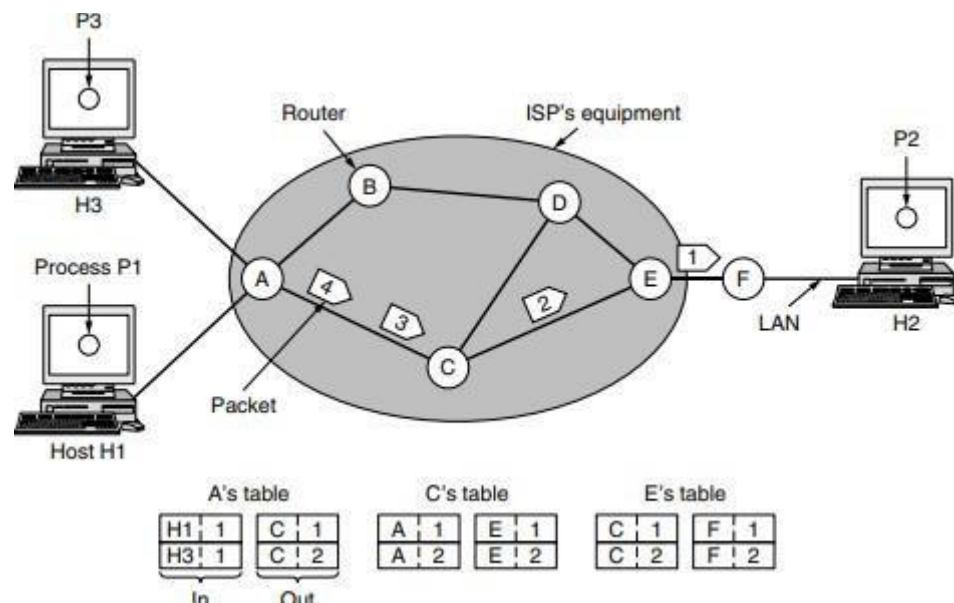
Having looked at the two classes of service the network layer can provide to its users, it is time to see how this layer works inside. Two different organizations are possible, depending on the type of service offered. If connectionless service is offered, packets are injected into the network individually and routed independently of each other. No advance setup is needed. In this context, the packets are frequently called datagram's (in analogy with telegrams) and the network is called a datagram network. If connection-oriented service is used, a path from the source router all the way to the destination router must be established before any data packets can be sent. This connection is called a VC (virtual circuit), in analogy with the physical circuits set up by the telephone system, and the network is called a virtual- circuit network. In this section, we will examine datagram networks; in the next one, we will examine virtual-circuit networks.

Let us now see how a datagram network works. Suppose that the process P1 in Fig. has a long message for P2. It hands the message to the transport layer, with instructions to deliver it to process P2 on host H2. The transport layer code runs on H1, typically within the operating system. It prepends a transport header to the front of the message and hands the result to the network layer, probably just another procedure within the operating system



4 . Implementation of connection-oriented service

For connection-oriented service, we need a virtual-circuit network. Let us see how that works. The idea behind virtual circuits is to avoid having to choose a new route for every packet sent, as in Fig. 5-2. Instead, when a connection is established, a route from the source machine to the destination machine is chosen as part of the connection setup and stored in tables inside the routers. That route is used for all traffic flowing over the connection, exactly the same way that the telephone system works. When the connection is released, the virtual circuit is also terminated. With connection-oriented service, each packet carries an identifier telling which virtual circuit it belongs to. As an example, consider the situation shown in Fig. 5-3. Here, host H1 has established connection 1 with host H2. This connection is remembered as the first entry in each of the routing tables. The first line of A's table says that if a packet bearing connection identifier 1 comes in from H1, it is to be sent to router C and given connection identifier 1. Similarly, the first entry at C routes the packet to E, also with connection identifier 1.



Now let us consider what happens if H3 also wants to establish a connection to H2. It chooses connection identifier 1 (because it is initiating the connection and this is its only connection) and tells the network to establish the virtual circuit. This leads to the second row in the tables. Note that we have a conflict here because although A can easily distinguish connection 1 packets from H1 from connection 1 packets from H3, C cannot do this. For this reason, A assigns a different connection identifier to the outgoing traffic for the second connection. Avoiding conflicts of this kind is why routers need the ability to replace connection identifiers in outgoing packets.

5. Comparison of Virtual Circuit Subnets and Datagram Subnets:

Issue	Datagram network	Virtual-circuit network
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
State information	Routers do not hold state information about connections	Each VC requires router table space per connection
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated
Quality of service	Difficult	Easy if enough resources can be allocated in advance for each VC
Congestion control	Difficult	Easy if enough resources can be allocated in advance for each VC

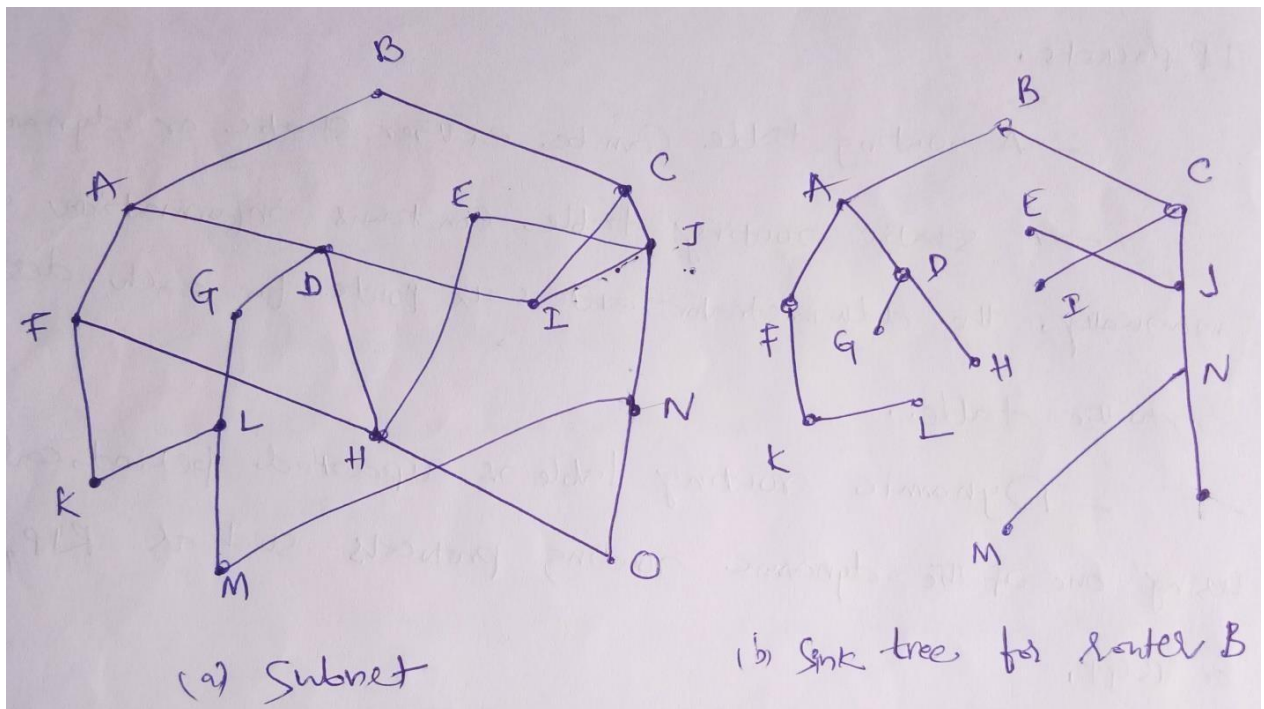
Routings Algorithm – Optimal Principle:

A host or a router has a routing table with an entry for each destination, or a combination of destinations, to route IP packets. A routing table can be either static or dynamic.

The main function of the network layer is to route packets from source to destination, to accomplish this on route through the network must be selected, and generally more than one route is possible.

Optimality refers to the ability of the routing algorithm to select the best route and the best route depends on the metrics and these metric weightings used to make the calculation.

For example, one routing algorithm might delay more heavily in the calculation. The figure shows that principle of optimality



Optimality principle states that if router J is on the optimal path from router I to router K, then the optimal path from J to K also falls, along the same route. Suppose route from I to J is R_1 and rest of the route is called R_2 .

If a route better than R_2 is existed from J to K, it could be concatenated with R_1 to improve the route from I to K. So that $R_1 R_2$ is optimal.

From an above figure shows the subnet and sink tree with distance metric is measured as the number of hops. Sink tree is not necessarily unique; other trees with the same path lengths may exist. Sink tree does not contain any loops so each packet will be delivered within a finite and bounded number of hops.

Shortest Path Routing or Dijkstra's Algorithm:

In shortest path routing the path length between each node is measured as a function of distance, bandwidth, average traffic, communication cost, mean queue length, measured delay etc.

By changing the weighting function, the algorithm then computes the shortest path measured according to any one of a number of criteria or a combination of criteria.

For this we draw a graph of subnet. In this each node of graph representing a router and each arc of the graph representing a communication link. Each link has a cost associated with it.

To choose a route between a given pair of routers, the algorithm just finds the shortest path between there on the graph. The algorithm for computing the shortest path between two nodes of a graph are called Dijkstra's algorithm

Dijkstra's Algorithm:

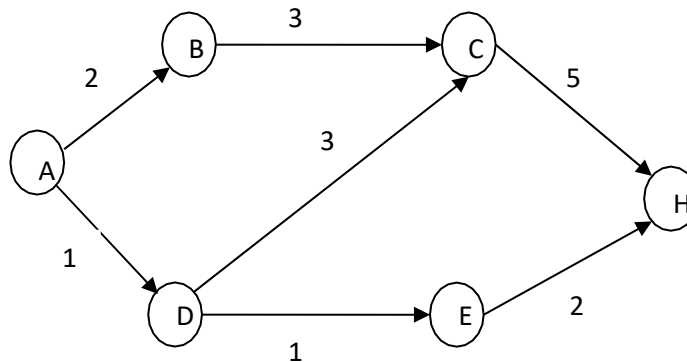
In Dijkstra's algorithm each node is labelled with its distance from the source node along the best-known path.

Initially no paths are known, so all nodes are labelled with infinity (∞). As the algorithm proceeds, the paths are found, the labels are changed, reflecting better paths,

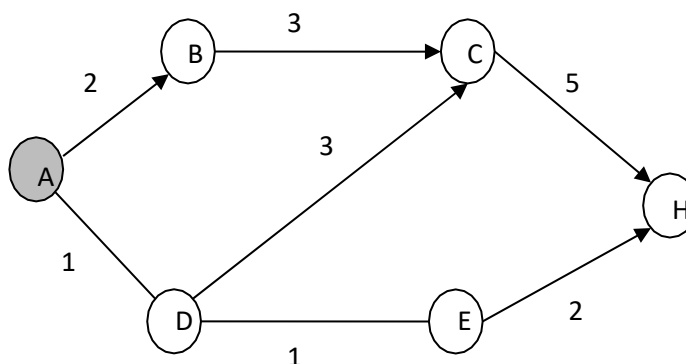
The algorithm follows as step by step

Source node is initialized and can be indicated as a solid circle.

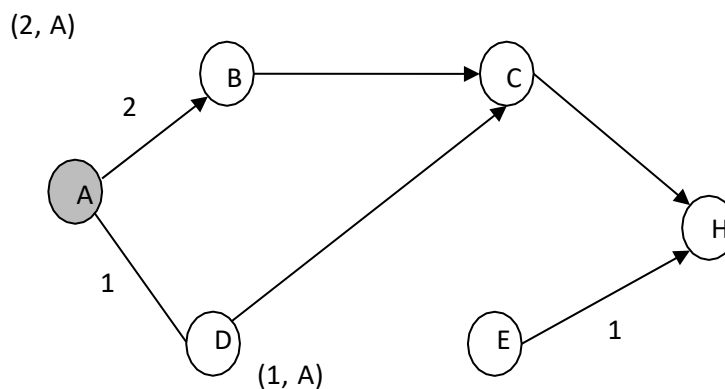
1. Initial path cost to neighbouring nodes (adjacent nodes) or link cost is computed and these nodes are relabelled considering source node.
2. Examine the all adjacent nodes and find the smallest label, make it permanent.
3. The smallest label node is now working node, then step 2 & step 3 are repeated till the destination node reached



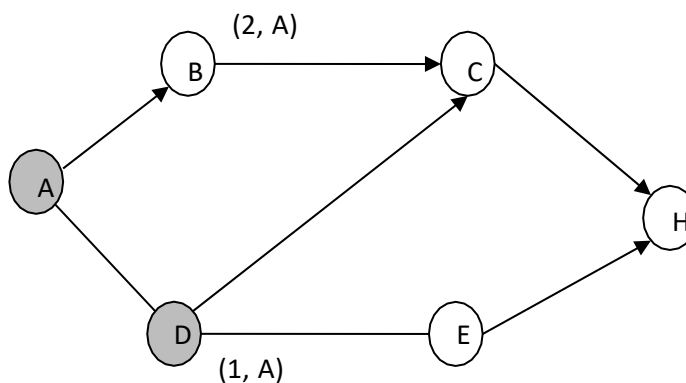
Step 1: node A is initialized as source node



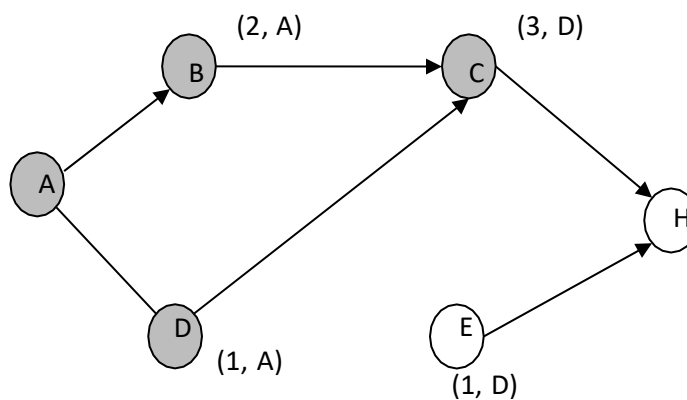
Step 2: link cost is computed for the adjacent node.



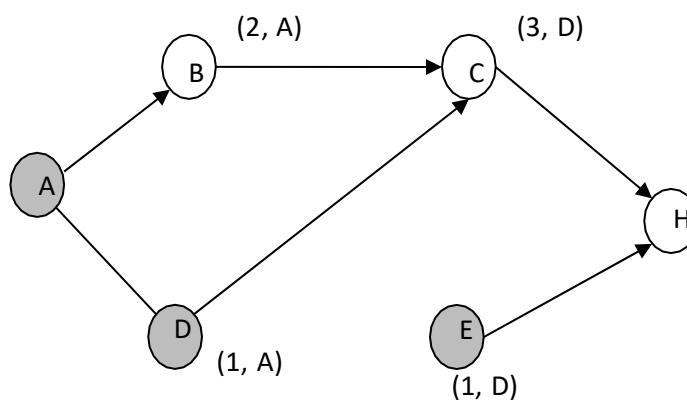
Step 3: Since AD is the smallest path, now D is working node.



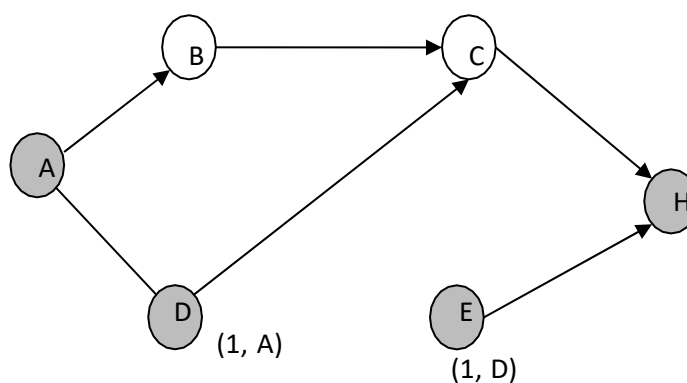
Step 4: adjacent nodes to D are C & E



Step 5: since shortest is E, now E is working node.



Step 6:



Hence the shortest path between node A & node H is ADEH

Flooding:

The flooding technique requires no network information. A packet is sent by a source node to all its adjacent nodes. At each node, every incoming packet is retransmitted on every outgoing links, except the link that it arrived from.

Flooding generates large no. of duplicate packets.

One way to prevent this is for each node to renumber the identity of those packets it has already sent. When duplicate packets arrive they are discarded.

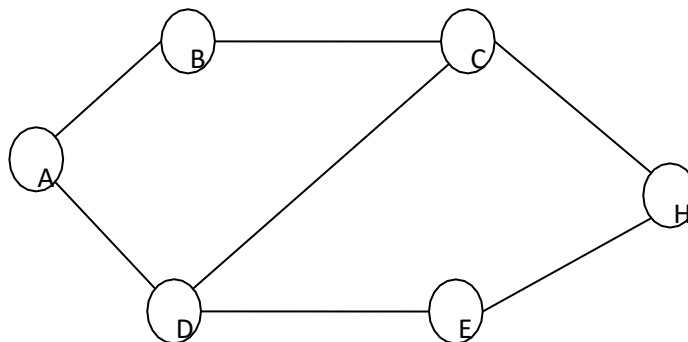
Another measure is to have a hop counter contained in the header of each packet, which is decremented at each hop. When count reaches to zero, the packet is discarded the counter is set to maximum value i.e, diameter of the subnet.

In selective flooding the router do not retransmit every incoming packets on all links but only on those links that are in the right direction.

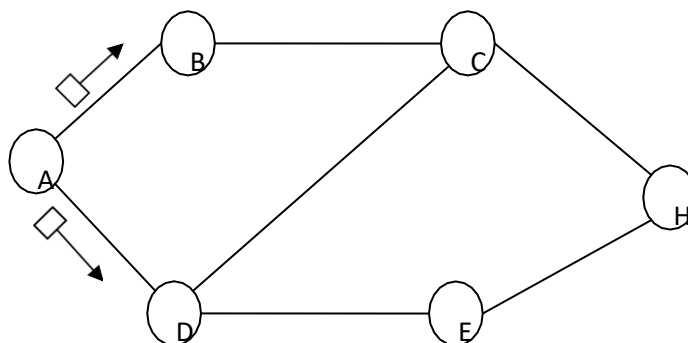
The main disadvantage of flooding is the total traffic load that is generates, is directly proportional to the connectivity of the network. Also flooding requires much large band width.

The advantage of flooding is that it is highly robust.

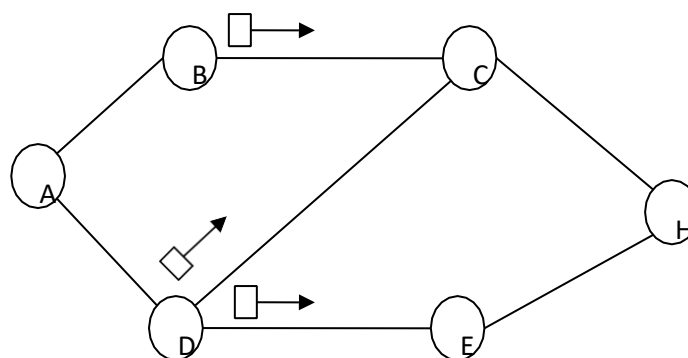
The following figures shows that flooding process the subnet is given as.



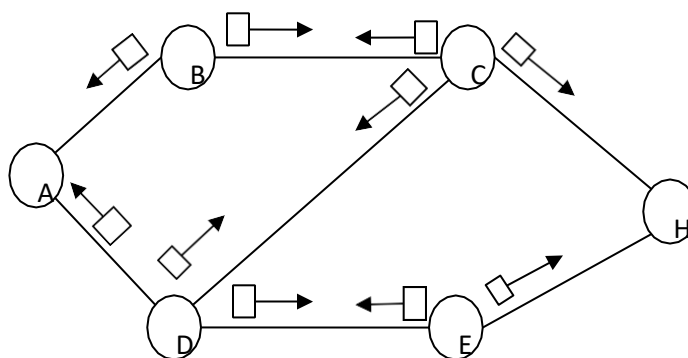
The first hop is:



The second hop is



The third hop
is:



Dynamic Routing Algorithms:

1. Distance Vector Routing:

A **distance vector routing** algorithm operates by having each router maintain a table (i.e., a vector) giving the best known distance to each destination and which link to use to get there. These tables are updated by exchanging information with the neighbors. Eventually, every router knows the best link to reach each destination. The distance vector routing algorithm is sometimes called by other names, most commonly the distributed **Bellman-Ford** routing algorithm, after the researchers who developed it. It's a dynamic routing algorithm. It was the original ARPANET routing algorithm and was also used in the Internet under the name RIP. In distance vector routing, each router maintains a routing table known as **vector**. This local routing table containing one entry for each router in the network. This entry has three parts: the destination, an estimate of the distance to that destination and next hop.

How it Works:

Step-1:

Each router prepares its routing table. By their local knowledge. Each router knows about-

- All the Routers present in the network.
- Distance to its neighbor Routers.

Step-2:

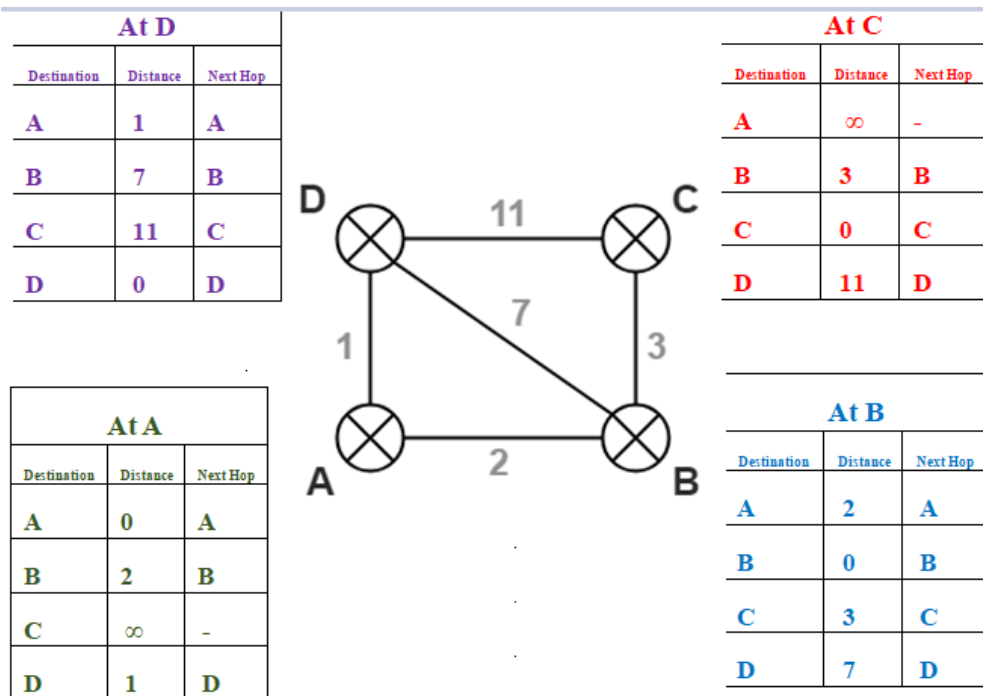
- Each router exchanges its distance vector with its neighboring routers.
- Each router prepares a new routing table using the distance vectors it has obtained from its neighbors.
- This step is repeated for (n-1) times if there are n routers in the network.
- After this, routing tables converge / become stable

Distance Vector Routing Example

Consider- There is a network consisting of 4 routers. The weights are mentioned on the edges. Weights could be distances or costs or delays.

Step-1:

Initially each router prepares its routing table using its local knowledge.

**Step: 2**

- Each router exchanges its distance vector obtained in Step-01 with its neighbors.
- After exchanging the distance vectors, each router prepares a new routing table.

At Router A

- Router A receives distance vectors from its neighbors B and D.
- Router A prepares a new routing table as-

From B	From D	
2	1	
0	7	
3	11	
7	0	

Destination	Distance	Next hop
A	0	A
B		
C		
D		

Cost(A→B) = 2

Cost(A→D) = 1

New Routing Table at Router A

- Cost of reaching destination B from router A = $\min \{ 2+0, 1+7 \} = 2$ via B.
- Cost of reaching destination C from router A = $\min \{ 2+3, 1+11 \} = 5$ via B.
- Cost of reaching destination D from router A = $\min \{ 2+7, 1+0 \} = 1$ via D

Explanation for Destination B

- Router A can reach the destination router B via its neighbor B or neighbor D.
- It chooses the path which gives the minimum cost.
- Cost of reaching router B from router A via neighbor B = Cost (A→B) + Cost (B→B) = $2 + 0 = 2$
- Cost of reaching router B from router A via neighbor D = Cost (A→D) + Cost (D→B) = $1 + 7 = 8$
- Since the cost is minimum via neighbor B, so router A chooses the path via B.
- It creates an entry (2, B) for destination B in its new routing table.
- Similarly, we calculate the shortest path distance to each destination router at every router.

Thus, the new routing table at router A is-

Destination	Distance	Next Hop
A	0	A
B	2	B
C	5	B
D	1	D

At Router B

- Router B receives distance vectors from its neighbors A, C and D.
- Router B prepares a new routing table as-

From A	From C	From D
0	∞	1
2	3	7
∞	0	11
1	11	0
Cost (B→A) = 2	Cost (B→C) = 3	Cost (B→D) = 7

- Cost of reaching destination A from router B = $\min \{ 2+0, 3+\infty, 7+1 \} = 2$ via A.
- Cost of reaching destination C from router B = $\min \{ 2+\infty, 3+0, 7+11 \} = 3$ via C.
- Cost of reaching destination D from router B = $\min \{ 2+1, 3+11, 7+0 \} = 3$ via A.

Thus, the new routing table at router B is-

Destination	Distance	Next Hop
A	2	A
B	0	B
C	3	C
D	3	A

At Router C-

- Router C receives distance vectors from its neighbors B and D.
- Router C prepares a new routing table as-

From B	From D
2	1
0	7
3	11
7	0

Cost (C→B) = 3 Cost (C→D) = 11

- Cost of reaching destination A from router C = $\min \{ 3+2, 11+1 \} = 5$ via B.
- Cost of reaching destination B from router C = $\min \{ 3+0, 11+7 \} = 3$ via B.
- Cost of reaching destination D from router C = $\min \{ 3+7, 11+0 \} = 10$ via B.

Thus, the new routing table at router C is

Destination	Distance	Next Hop
A	5	B
B	3	B
C	0	C
D	10	B

At Router D-

- Router D receives distance vectors from its neighbors A, B and C.
- Router D prepares a new routing table as-

From A	From B	From C
0	2	∞
2	0	3
∞	3	0
1	7	11

Cost (D→A) = 1 Cost (D→B) = 7 Cost (D→C) = 11

- Cost of reaching destination A from router D = $\min \{ 1+0, 7+2, 11+\infty \} = 1$ via A.

- Cost of reaching destination B from router D = $\min \{ 1+2, 7+0, 11+3 \} = 3$ via A.
- Cost of reaching destination C from router D = $\min \{ 1+\infty, 7+3, 11+0 \} = 10$ via B.

Thus, the new routing table at router D is

Destination	Distance	Next Hop
A	1	A
B	3	A
C	10	B
D	0	D

Step-03:

- Each router exchanges its distance vector obtained in Step-02 with its neighboring routers.
- After exchanging the distance vectors, each router prepares a new routing table.

At Router A

- Router A receives distance vectors from its neighbors B and D.
- Router A prepares a new routing table as-

From B

2
0
3
3

From D

1
3
10
0

Cost(A→B) = 2 Cost(A→D) = 1

- Cost of reaching destination B from router A = $\min \{ 2+0, 1+3 \} = 2$ via B.
- Cost of reaching destination C from router A = $\min \{ 2+3, 1+10 \} = 5$ via B.
- Cost of reaching destination D from router A = $\min \{ 2+3, 1+0 \} = 1$ via D.

Thus, the new routing table at router A is-

Destination	Distance	Next Hop
A	0	A
B	2	B
C	5	B
D	1	D

At Router B-

- Router B receives distance vectors from its neighbors A, C and D.
- Router B prepares a new routing table as-

From A	From C	From D
0	5	1
2	3	3
5	0	10
1	10	0

Cost (B→A) = 2 Cost (B→C) = 3 Cost (B→D) = 3

- Cost of reaching destination A from router B = $\min \{ 2+0, 3+5, 3+1 \} = 2$ via A.
- Cost of reaching destination C from router B = $\min \{ 2+5, 3+0, 3+10 \} = 3$ via C.
- Cost of reaching destination D from router B = $\min \{ 2+1, 3+10, 3+0 \} = 3$ via A.

Thus, the new routing table at router B is-

Destination	Distance	Next Hop
A	2	A
B	0	B
C	3	C
D	3	A

At Router C-

- Router C receives distance vectors from its neighbors B and D.
- Router C prepares a new routing table as-

From B	From D
2	1
0	3
3	10
3	0

Cost (C→B) = 3 Cost (C→D) = 10

- Cost of reaching destination A from router C = $\min \{ 3+2, 10+1 \} = 5$ via B.
- Cost of reaching destination B from router C = $\min \{ 3+0, 10+3 \} = 3$ via B.
- Cost of reaching destination D from router C = $\min \{ 3+3, 10+0 \} = 6$ via B.

Thus, the new routing table at router C is-

Destination	Distance	Next Hop
A	5	B
B	3	B
C	0	C
D	6	B

At Router D-

- Router D receives distance vectors from its neighbors A, B and C.
- Router D prepares a new routing table as-

From A	From B	From C
0	2	5
2	0	3
5	3	0
1	3	10

Cost (D→A) = 1 Cost (D→B) = 3 Cost (D→C) = 10

- Cost of reaching destination A from router D = $\min \{ 1+0, 3+2, 10+5 \} = 1$ via A.
- Cost of reaching destination B from router D = $\min \{ 1+2, 3+0, 10+3 \} = 3$ via A.
- Cost of reaching destination C from router D = $\min \{ 1+5, 3+3, 10+0 \} = 6$ via A.

Thus, the new routing table at router D is-

Destination	Distance	Next Hop
A	1	A
B	3	A
C	6	A
D	0	D

This is how final routing tables will be updated at each router.

This updating process is illustrated in Fig. 5-9. Part (a) shows a network. The first four columns of part (b) show the delay vectors received from the neighbours of router J. A claims to have a 12-msec delay to B, a 25-msec delay to C, a 40- msec delay to D, etc. Suppose that J has measured or estimated its delay to its neighbours, A, I, H, and K, as 8, 10, 12, and 6 msec, respectively.

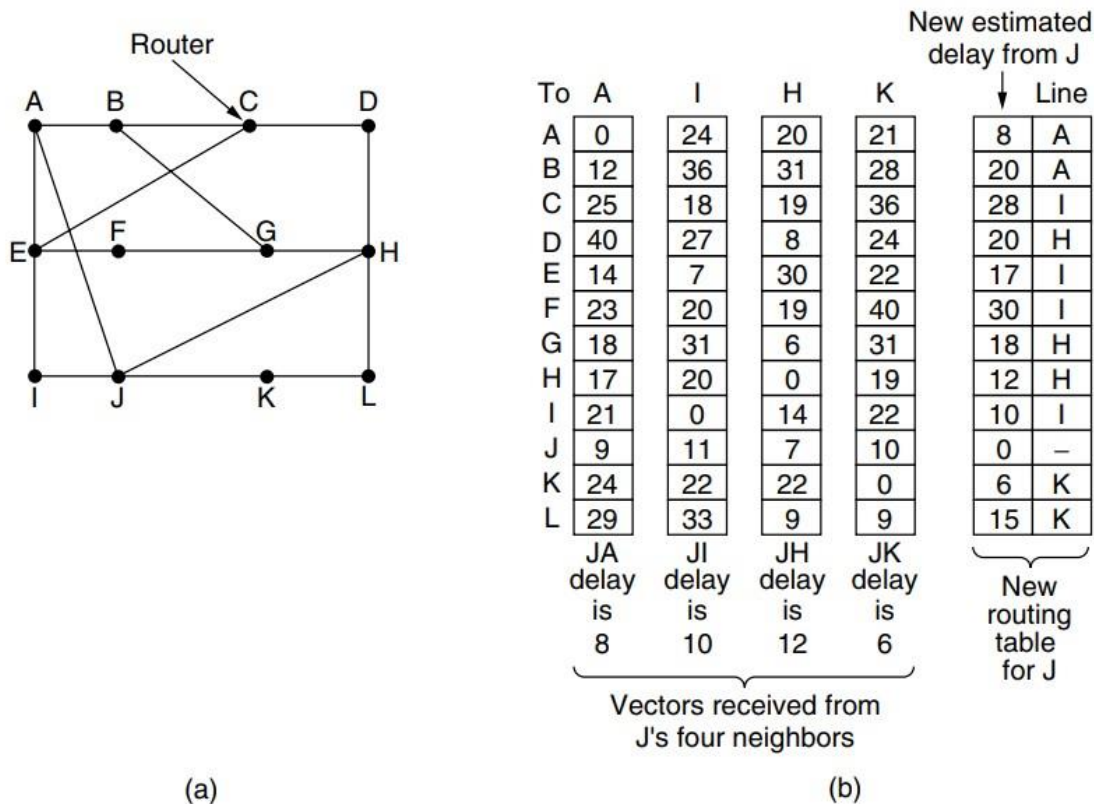


Figure 5-9. (a) A network. (b) Input from A, I, H, K, and the new routing table for J.

Consider how J computes its new route to router G. It knows that it can get to A in 8 msec, and furthermore A claims to be able to get to G in 18 msec, so J knows it can count on a delay of 26 msec to G if it forwards packets bound for G to A. Similarly, it computes the delay to G via I, H, and K as 41 ($31 + 10$), 18 ($6 + 12$), and 37 ($31 + 6$) msec, respectively. The best of these values is 18, so it makes an entry in its routing table that the delay to G is 18 msec and that the route to use is via H. The same calculation is performed for all the other destinations, with the new routing table shown in the last column of the figure.

Count – To Infinity Problem:

The count – to – infinity problem occurs when the subnet is given as In the above shows an imagined network and denotes the distances from router A to every other router.

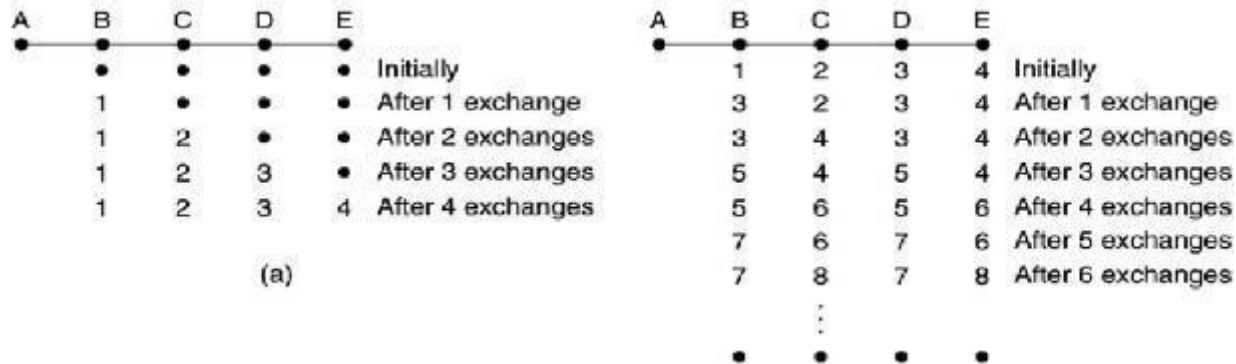
Suppose that link (A,B) fails (or) broken router B observed it, but in routing table of B, C has a router to a with 2 hops

The problem is that B does not know that C has a router B as Successor in routing table on the route to A. this followed count – to – infinity problem.

Router B actualizes own routing table and takes the router to A over router C. we can see that new distances to A.

In router C's routing the router to A contains router B as next hop router, so if B has increased his costs to A, C is forced to do so. Router C increase his cost to A about $B+1 = 4$

Now we see the consequence of the distributed Bellman ford protocol.



Router B takes the path over C to A, and re actualizes own routing table and So on. There are several partial solutions to the count – to – infinity problem

1. Some relatively small number as an approximation of infinity

Example: we may decide that the maximum no. of hops to get across a certain. Network is never going to be more than 16 and so we could pick 16 as the value that represents infinity. This bounds the amount of time that it takes to count to infinity of course it could also present a problem if our network grew to a point where some nodes were separated by more than 16 hops

2. Another is to improve the time to stabilize routing is called split horizon.

This technique implies that routing information about some network stored in the routing table of a specific router is never sent to the router from which it was received.

2. Link State Routing

Distance vector routing was used in the ARPANET until 1979, when it was replaced by link state routing. The primary problem that caused its demise was that the algorithm often took too long to converge after the network topology changed (due to the count-to-infinity problem). Consequently, it was replaced by an entirely new algorithm, now called link state routing. Variants of link state routing called IS-IS and OSPF are the routing algorithms that are most widely used inside large networks and

the Internet today. The idea behind link state routing is fairly simple and can be stated as five parts. Each router must do the following things to make it work:

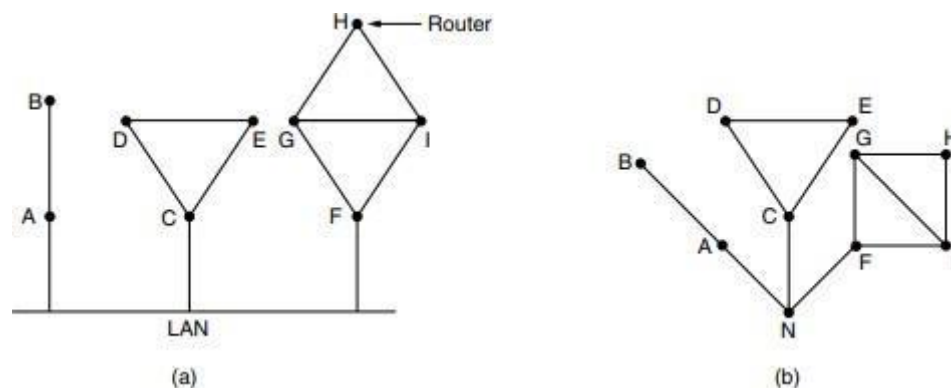
1. Discover its neighbours and learn their network addresses.
2. Set the distance or cost metric to each of its neighbours.
3. Construct a packet telling all it has just learned.
4. Send this packet to and receive packets from all other routers.
5. Compute the shortest path to every other router.

In effect, the complete topology is distributed to every router. Then Dijkstra's algorithm can be run at each router to find the shortest path to every other router. Below we will consider each of these five steps in more detail.

i. Learning about the Neighbors

When a router is booted, its first task is to learn who its neighbors are. It accomplishes this goal by sending a special HELLO packet on each point-to-point line. The router on the other end is expected to send back a reply giving its name. These names must be globally unique because when a distant router later hears that three routers are all connected to F, it is essential that it can determine whether all three mean the same F.

When two or more routers are connected by a broadcast link (e.g., a switch, ring, or classic Ethernet), the situation is slightly more complicated. Fig. (a) illustrates a broadcast LAN to which three routers, A, C, and F, are directly connected. Each of these routers is connected to one or more additional routers, as shown.



ii. Setting Link Costs

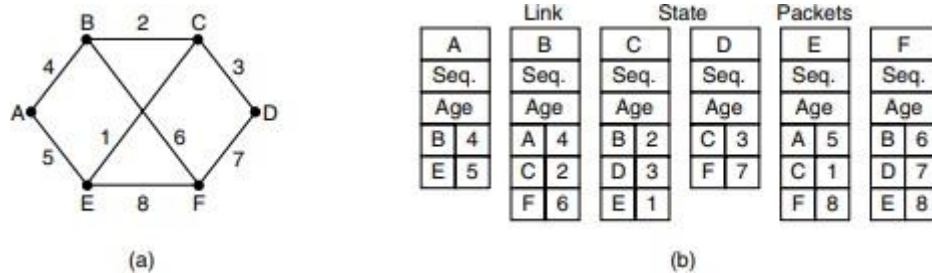
The link state routing algorithm requires each link to have a distance or cost metric for finding shortest paths. The cost to reach neighbors can be set automatically, or configured by the network operator. A common choice is to make the cost inversely proportional to the bandwidth of the link. For example, 1-Gbps Ethernet may have a cost of 1 and 100-Mbps Ethernet a cost of 10. This makes higher-capacity paths better choices.

If the network is geographically spread out, the delay of the links may be factored into the cost so that paths over shorter links are better choices. The most direct way to determine this delay is to send over the line a special ECHO packet that the other side is required to send back immediately. By measuring the round-trip time and dividing it by two, the sending router can get a reasonable estimate of the delay.

iii. Building Link State Packets

Once the information needed for the exchange has been collected, the next step is for each router to build a packet containing all the data. The packet starts with the identity of the sender, followed by a sequence

number and age (to be described later) and a list of neighbors. The cost to each neighbor is also given. An example network is presented in Fig. (a) with costs shown as labels on the lines. The corresponding link state packets for all six routers are shown in Fig.(b).



iv. Distributing the Link State Packets

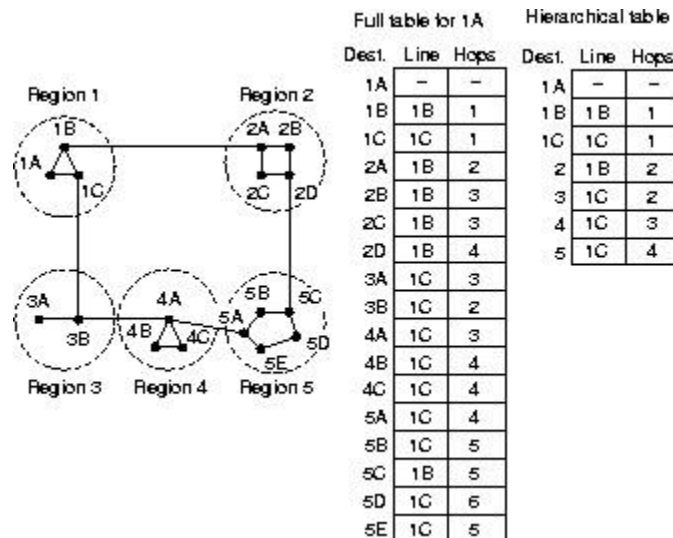
The trickiest part of the algorithm is distributing the link state packets. All of the routers must get all of the link state packets quickly and reliably. If different routers are using different versions of the topology, the routes they compute can have inconsistencies such as loops, unreachable machines, and other problems.

Source	Seq.	Age	Send flags			ACK flags			Data
			A	C	F	A	C	F	
A	21	60	0	1	1	1	0	0	
F	21	60	1	1	0	0	0	1	
E	21	59	0	1	0	1	0	1	
C	20	60	1	0	1	0	1	0	
D	21	59	1	0	0	0	1	1	

v. Computing the New Routes

Now Dijkstra's algorithm can be run locally to construct the shortest paths to all possible destinations. The results of this algorithm tell the router which link to use to reach each destination. This information is installed in the routing tables, and normal operation is resumed.

3. Hierarchical Routing



From an above figure, At a certain point the network may grow to the point where it is no longer feasible for every router to have an entry for other router, so the routing will have to be done hierarchically

When this routing is used, the routers are divided into regions. It contains all the details about how to route packets to destination within its own region.

Some time, two – level hierarchy may be in – efficient it is necessary to group the regions into clusters the clusters into zones and zones into groups and so on.

The above figure shows routing in a two level hierarchy.

Hierarchy For Table 1A:

<i>Destination</i>	<i>Line</i>	<i>Hops</i>
<i>1A</i>	-	-
<i>1B</i>	<i>1B</i>	<i>1</i>
<i>1C</i>	<i>1C</i>	<i>1</i>

2	1B	2
3	1C	2
4	1C	3
5	1C	5

Full table for 1A

<i>Destination</i>	<i>Line</i>	<i>Hops</i>
1A	-	-
1B	1B	1
1C	1C	1
2A	1B	2
2B	1B	3
2C	1B	3
2D	1B	4
3A	1C	3
3B	1C	2
4A	1C	3
4B	1C	4
4C	1C	4
5A	1C	5
5B	1C	6
5C	1B	5
5D	1C	7
5E	1C	6

From an above routing table, for router 1A has 17 entries in full routing table. When hierarchical routing is used only 07 entries valid for 1A

There are entries for local routers but all other regions have been condensed into a signal router.

Hierarchical routing increases the saving the table space.

Broadcast Routing

In some applications, hosts need to send messages to many or all other hosts. For example, a service distributing weather reports, stock market updates, or live radio programs might work best by sending to all machines and letting those that are interested read the data. Sending a packet to all destinations simultaneously is called broadcasting.

Various methods have been proposed for doing it.

One broadcasting method that requires no special features from the network is for the source to simply send a distinct packet to each destination. Not only is the method wasteful of bandwidth and slow, but it also requires the source to have a complete list of all destinations. This method is not desirable in practice, even though it is widely applicable.

An improvement is multi-destination routing, in which each packet contains either a list of destinations or a bit map indicating the desired destinations. When a packet arrives at a router, the router checks all the destinations to determine the set of output lines that will be needed. (An output line is needed if it is the best route to at least one of the destinations.) The router generates a new copy of the packet for each output line to be used and includes in each packet only those destinations that are to use the line. In effect, the destination set is partitioned among the output lines. After a sufficient number of hops, each packet will carry only one destination like a normal packet. Multi- destination routing is like using separately addressed packets, except that when several packets must follow the same route, one of them pays full fare and the rest ride free. The network bandwidth is therefore used more efficiently. However, this scheme still requires the source to know all the destinations, plus it is as much work for a router to determine where to send one multi-destination packet as it is for multiple distinct packets.

We have already seen a better broadcast routing technique: flooding. When implemented with a sequence number per source, flooding uses links efficiently with a decision rule at routers that is relatively simple. Although flooding is ill-suited for ordinary point-to-point communication, it rates serious consideration for broadcasting. However, it turns out that we can do better still once the shortest path routes for regular packets have been computed.

The idea for reverse path forwarding is elegant and remarkably simple once it has been pointed out. When a broadcast packet arrives at a router, the router checks to see if the packet arrived on the link that is normally used for sending packets toward the source of the broadcast. If so, there is an excellent chance that the broadcast packet itself followed the best route from the router and is therefore the first copy to arrive at the router. This being the case, the router forwards copies of it onto all links except the one it arrived on. If, however, the broadcast packet arrived on a link other than the preferred one for reaching the source, the packet is discarded as a likely duplicate.

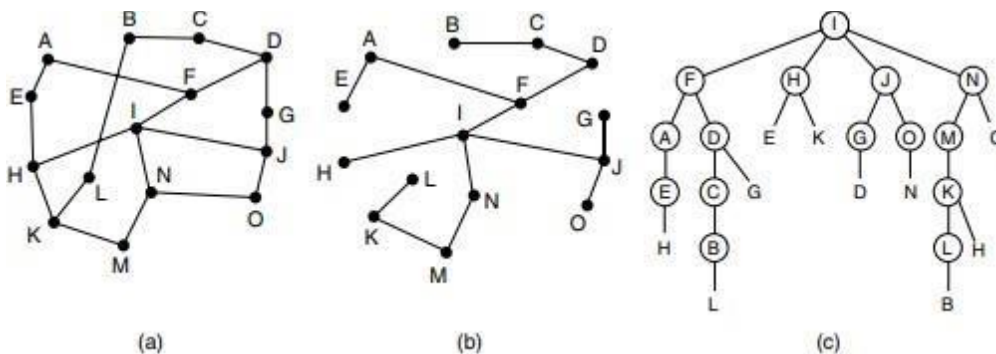


Fig. Reverse path forwarding. (a) A network. (b) A sink tree. (c) The tree built by reverse path forwarding.

An example of reverse path forwarding is shown in Fig. 5-15. Part (a) shows a network, part (b) shows a sink tree for router I of that network, and part (c) shows how the reverse path algorithm works. On the first hop, I sends packets to F, H, J, and N, as indicated by the second row of the tree. Each of these packets arrives on the preferred path to I (assuming that the preferred path falls along the sink tree) and is so indicated by a circle around the letter. On the second hop eight packets are generated, two by each of the routers that received a packet on the first hop. As it turns out, all eight of these arrive at previously unvisited routers, and five of these arrive along the preferred line. Of the six packets generated on the third hop, only three arrive on the preferred path (at C, E, and K); the others are duplicates. After five hops and 24 packets, the broadcasting terminates, compared with four hops and 14 packets had the sink tree been followed exactly.

The principal advantage of reverse path forwarding is that it is efficient while being easy to implement. It sends the broadcast packet over each link only once in each direction, just as in flooding, yet it requires only that routers know how to reach all destinations, without needing to remember sequence numbers (or use other mechanisms to stop the flood) or list all destinations in the packet.

Multicast Routing:

Some applications, such as a multiplayer game or live video of a sports event streamed to many viewing locations, send packets to multiple receivers. Unless the group is very small, sending a distinct packet to each receiver is expensive. On the other hand, broadcasting a packet is wasteful if the group consists of, say, 1000 machines on a million-node network, so that most receivers are not interested in the message (or worse yet, they are definitely interested but are not supposed to see it). Thus, we need a way to send messages to well-defined groups that are numerically large in size but small compared to the network as a whole.

Sending a message to such a group is called multicasting, and the routing algorithm used is called multicast routing. All multicasting schemes require some way to create and destroy groups and to identify which routers are members of a group.

As an example, consider the two groups, 1 and 2, in the network shown in Fig. 5-16(a). Some routers are attached to hosts that belong to one or both of these groups, as indicated in the figure. A spanning tree for the leftmost router is shown in Fig. 5-16(b). This tree can be used for broadcast but is overkill for multicast, as can be seen from the two pruned versions that are shown next. In Fig. 5-16(c), all the links that do not lead to hosts that are members of group 1 have been removed. The result is the multicast spanning tree for the leftmost router to send to group 1. Packets are forwarded only along this spanning tree, which is more efficient than the broadcast tree because there are 7 links instead of 10. Fig. 5-16(d) shows the multicast spanning tree after pruning for group 2. It is efficient too, with only five links this time. It also shows that different multicast groups have different spanning trees.

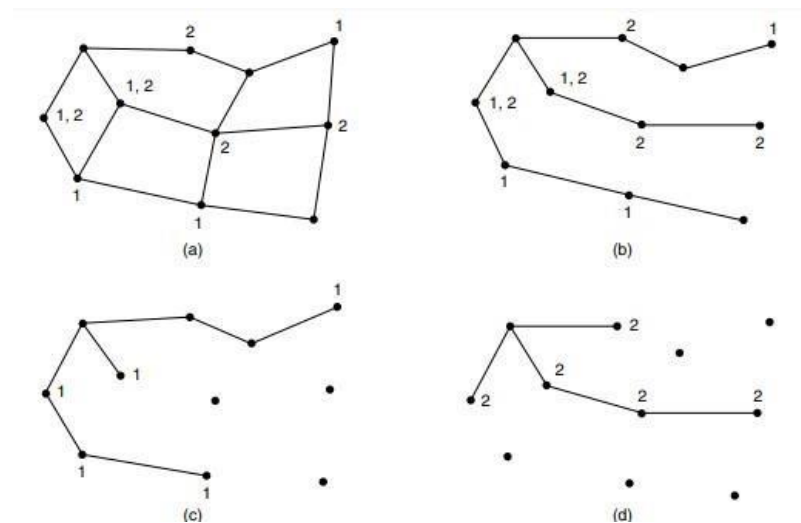
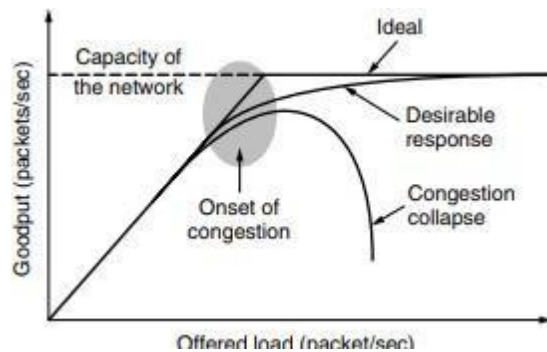


Fig. (a) A network. (b) A spanning tree for the leftmost router. (c) A multicast tree for group 1. (d) A multicast tree for group 2.

Congestion is an important issue that can arise in packet switched network. Congestion is a situation in Communication Networks in which too many packets are present in a part of the subnet, performance degrades. Congestion in a network may occur when the load on the network (*i.e.* the number of packets sent to the network) is greater than the capacity of the network (*i.e.* the number of packets a network can handle.). Network congestion occurs in case of traffic overloading.

In other words when too much traffic is offered, congestion sets in and performance degrades sharply.

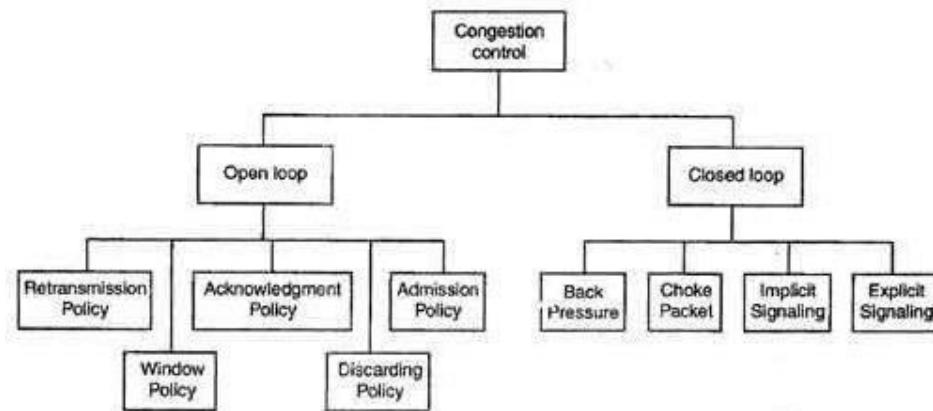


The various causes of congestion in a subnet are:

- The input traffic rate exceeds the capacity of the output lines. If suddenly, a stream of packet start arriving on three or four input lines and all need the same output line. In this case, a queue will be built up. If there is insufficient memory to hold all the packets, the packet will be lost. Increasing the memory to unlimited size does not solve the problem. This is because, by the time packets reach front of the queue, they have already timed out (as they waited the queue). When timer goes off source transmits duplicate packet that are also added to the queue. Thus same packets are added again and again, increasing the load all the way to the destination.
- The routers are too slow to perform bookkeeping tasks (queuing buffers, updating tables, etc.).
- The routers buffer is too limited.
- Congestion in a subnet can occur if the processors are slow. Slow speed CPU at routers will perform the routine tasks such as queuing buffers, updating table etc slowly. As a result of this, queues are built up even though there is excess line capacity.
- Congestion is also caused by slow links.

Congestion Control refers to techniques and mechanisms that can either prevent congestion, before it happens, or remove congestion, after it has happened. Congestion control mechanisms are divided into two categories,

one category prevents the congestion from happening and the other category removes congestion after it has taken place.



These two categories are:

1. Open loop
2. Closed loop

1. Open Loop Congestion Control

- In this method, policies are used to prevent the congestion before it happens.
- Congestion control is handled either by the source or by the destination.
- The various methods used for open loop congestion control are:

i. Retransmission Policy

- The sender retransmits a packet, if it feels that the packet it has sent is lost or corrupted.
- However retransmission in general may increase the congestion in the network. But we need to implement good retransmission policy to prevent congestion.
- The retransmission policy and the retransmission timers need to be designed to optimize efficiency and at the same time prevent the congestion.

ii. Window Policy

- To implement window policy, selective reject window method is used for congestion control.
- Selective Reject method is preferred over Go-back-n window as in Go-back-n method, when timer for a packet times out, several packets are resent, although some may have arrived safely at the receiver. Thus, this duplication may make congestion worse.
- Selective reject method sends only the specific lost or damaged packets.

iii. Acknowledgement Policy

- The acknowledgement policy imposed by the receiver may also affect congestion.
- If the receiver does not acknowledge every packet it receives it may slow down the sender and help prevent congestion.
- Acknowledgments also add to the traffic load on the network. Thus, by sending fewer acknowledgements we can reduce load on the network.
- To implement it, several approaches can be used:
 1. A receiver may send an acknowledgement only if it has a packet to be sent.
 2. A receiver may send an acknowledgement when a timer expires.
 3. A receiver may also decide to acknowledge only N packets at a time.

iv. Discarding Policy

- A router may discard less sensitive packets when congestion is likely to happen.
- Such a discarding policy may prevent congestion and at the same time may not harm the integrity of the transmission.

v. Admission Policy

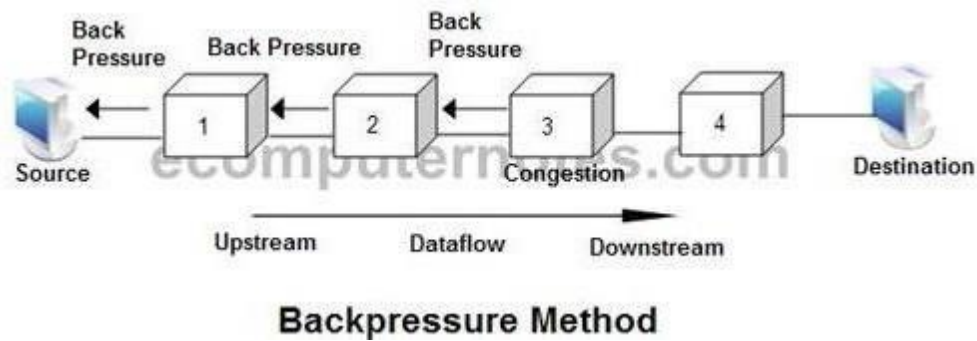
- An admission policy, which is a quality-of-service mechanism, can also prevent congestion in virtual circuit networks.
- Switches in a flow first check the resource requirement of a flow before admitting it to the network.
- A router can deny establishing a virtual circuit connection if there is congestion in the “network or if there is a possibility of future congestion.

2. Closed Loop Congestion Control

- Closed loop congestion control mechanisms try to remove the congestion after it happens.
- The various methods used for closed loop congestion control are:

1. Backpressure

Back pressure is a node-to-node congestion control that starts with a node and propagates, in the opposite direction of data flow.



The backpressure technique can be applied only to virtual circuit networks. In such virtual circuit each node knows the upstream node from which a data flow is coming.

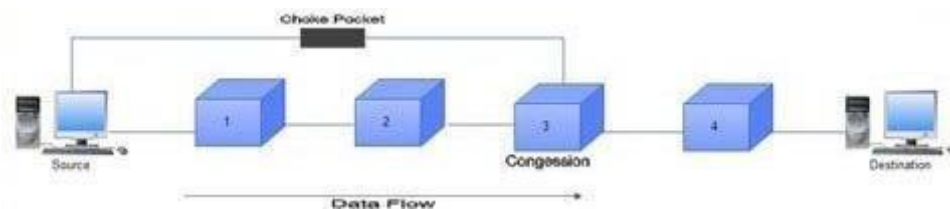
In this method of congestion control, the congested node stops receiving data from the immediate upstream node or nodes.

- This may cause the upstream node or nodes to become congested, and they, in turn, reject data from their upstream node or nodes.
- As shown in fig node 3 is congested and it stops receiving packets and informs its upstream node 2 to slow down. Node 2 in turn may be congested and informs node 1 to slow down. Now node 1 may create congestion and informs the source node to slow down. In this way the congestion is alleviated. Thus, the pressure on node 3 is moved backward to the source to remove the congestion.

2. Choke Packet

In this method of congestion control, congested router or node sends a special type of packet called choke packet to the source to inform it about the congestion.

- Here, congested node does not inform its upstream node about the congestion as in backpressure method.
- In choke packet method, congested node sends a warning directly to the source station *i.e.* the intermediate nodes through which the packet has travelled are not warned.



3. Implicit Signalling

- In implicit signalling, there is no communication between the congested node or nodes and the source.

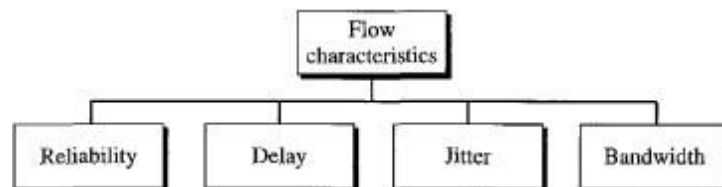
- The source guesses that there is congestion somewhere in the network when it does not receive any acknowledgment. Therefore the delay in receiving an acknowledgment is interpreted as congestion in the network.
- On sensing this congestion, the source slows down.
- This type of congestion control policy is used by TCP.

4. Explicit Signalling

- In this method, the congested nodes explicitly send a signal to the source or destination to inform about the congestion.
- Explicit signalling is different from the choke packet method. In choke packet method, a separate packet is used for this purpose whereas in explicit signalling method, the signal is included in the packets that carry data.
- Explicit signalling can occur in either the forward direction or the backward direction.
- In backward signalling, a bit is set in a packet moving in the direction opposite to the congestion. This bit warns the source about the congestion and informs the source to slow down.
- In forward signalling, a bit is set in a packet moving in the direction of congestion. This bit warns the destination about the congestion. The receiver in this case uses policies such as slowing down the acknowledgements to remove the congestion.

Quality of Service (QOS)

Traditionally, four types of characteristics are attributed to a flow: reliability, delay, jitter, and bandwidth.



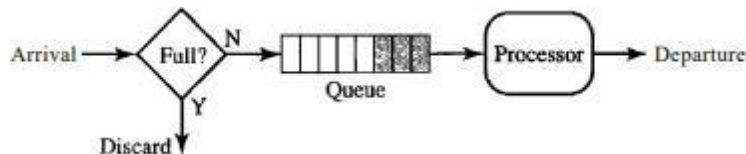
Techniques to improve QOS::

1. Scheduling:

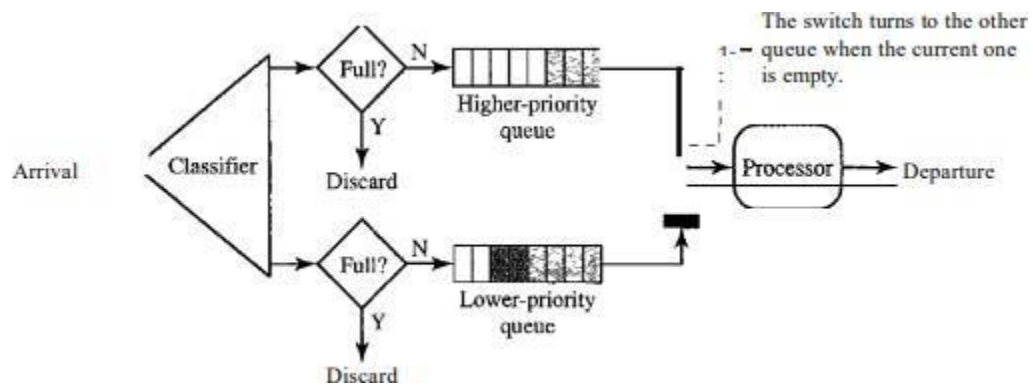
Scheduling Packets from different flows arrive at a switch or router for processing. A good scheduling technique treats the different flows in a fair and appropriate manner. Several scheduling techniques are

designed to improve the quality of service. We discuss three of them here: FIFO queuing, priority queuing, and weighted fair queuing.

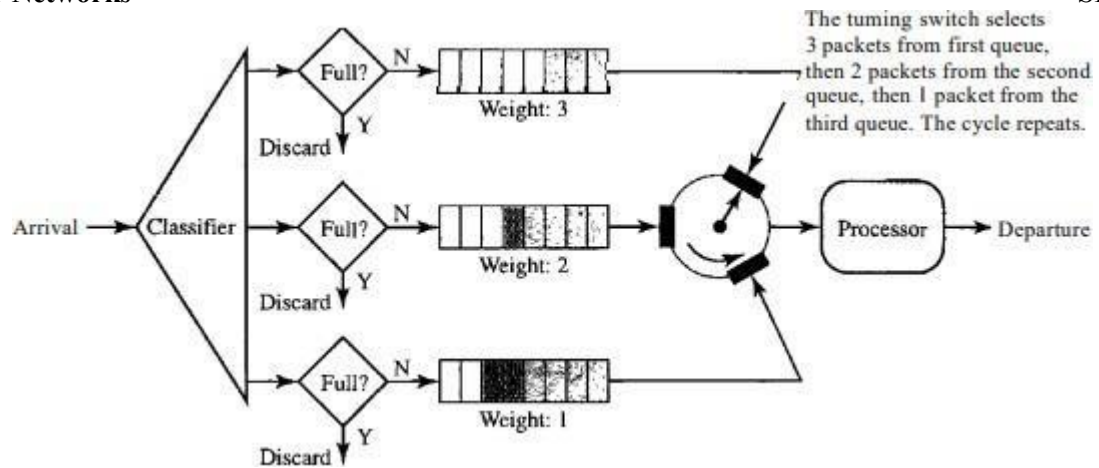
- i) **FIFO Queuing:** In first-in, first-out (FIFO) queuing, packets wait in a buffer (queue) until the node (router or switch) is ready to process them. If the average arrival rate is higher than the average processing rate, the queue will fill up and new packets will be discarded. Figure shows a conceptual view of a FIFO queue.



- i) **Priority Queuing:** In priority queuing, packets are first assigned to a priority class. Each priority class has its own queue. The packets in the highest-priority queue are processed first. Packets in the lowest-priority queue are processed last. Note that the system does not stop serving a queue until it is empty. Figure shows priority queuing with two priority levels (for simplicity)



- ii) **Weighted Fair Queuing** A better scheduling method is weighted fair queuing. In this technique, the packets are still assigned to different classes and admitted to different queues. The queues, however, are weighted based on the priority of the queues; higher priority means a higher weight. The system processes packets in each queue in a round-robin fashion with the number of packets selected from each queue based on the corresponding weight. For example, if the weights are 3, 2, and 1, three packets are processed from the first queue, two from the second queue, and one from the third queue. If the system does not impose priority on the classes, all weights can be equal. In this way, we have fair queuing with priority.

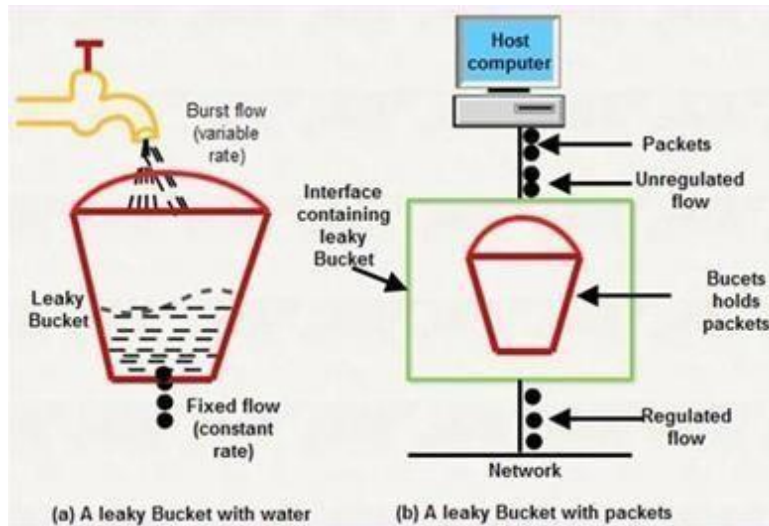


2. Traffic Shaping

Traffic shaping is a mechanism to control the amount and the rate of the traffic sent to the network. Two techniques can shape traffic: leaky bucket and token bucket.

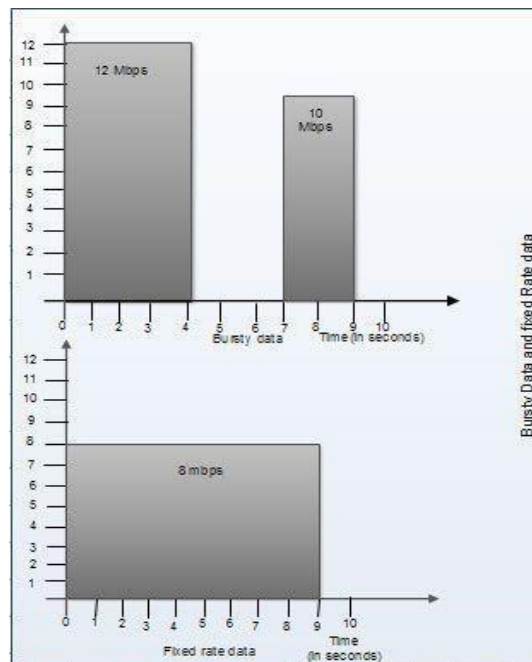
i) Leaky Bucket Algorithm

- It is a traffic shaping mechanism that controls the amount and the rate of the traffic sent to the network.
- A leaky bucket algorithm shapes bursty traffic into fixed rate traffic by averaging the data rate.
- Imagine a bucket with a small hole at the bottom.
- The rate at which the water is poured into the bucket is not fixed and can vary but it leaks from the bucket at a constant rate. Thus (as long as water is present in bucket), the rate at which the water leaks does not depend on the rate at which the water is input to the bucket.



- Also, when the bucket is full, any additional water that enters into the bucket spills over the sides and is lost.
- The same concept can be applied to packets in the network. Consider that data is coming from the source at variable speeds. Suppose that a source sends data at 12 Mbps for 4 seconds. Then there is no data for 3 seconds. The source again transmits data at a rate of 10 Mbps for 2 seconds. Thus, in a time span of 9 seconds, 68 Mb data has been transmitted.

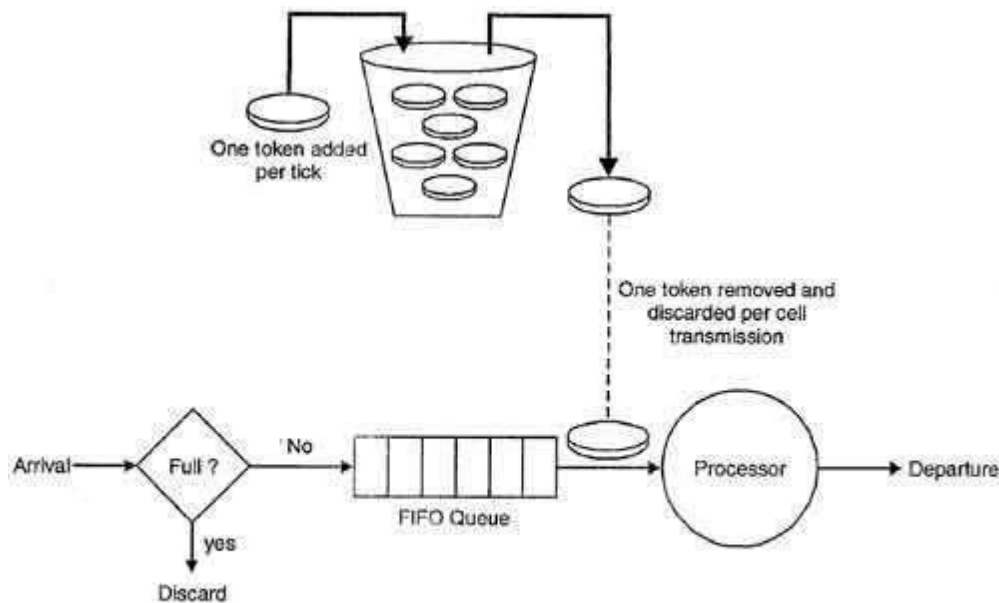
If a leaky bucket algorithm is used, the data flow will be 8 Mbps for 9 seconds. Thus constant flow is maintained.



ii) Token bucket Algorithm

- The leaky bucket algorithm allows only an average (constant) rate of data flow. Its major problem is that it cannot deal with bursty data.
- A leaky bucket algorithm does not consider the idle time of the host. For example, if the host was idle for 10 seconds and now it is willing to send data at a very high speed for another 10 seconds, the total data transmission will be divided into 20 seconds and average data rate will be maintained. The host is having no advantage of sitting idle for 10 seconds.
- To overcome this problem, a token bucket algorithm is used. A token bucket algorithm allows bursty data transfers.
- A token bucket algorithm is a modification of leaky bucket in which leaky bucket contains tokens.
- In this algorithm, a token(s) are generated at every clock tick. For a packet to be transmitted, system must remove token(s) from the bucket.
- Thus, a token bucket algorithm allows idle hosts to accumulate credit for the future in form of tokens.
- For example, if a system generates 100 tokens in one clock tick and the host is idle for 100 ticks. The bucket will contain 10,000 tokens.

Now, if the host wants to send bursty data, it can consume all 10,000 tokens at once for sending 10,000 cells or bytes. Thus a host can send bursty data as long as bucket is not empty.



Token bucket algorithm

- 3. Resource Reservation** A flow of data needs resources such as a buffer, bandwidth, CPU time, and so on. The quality of service is improved if these resources are reserved beforehand.
- 4. Admission Control** Admission control refers to the mechanism used by a router, or a switch, to accept or reject a flow based on predefined parameters called flow specifications. Before a router accepts a flow for processing, it checks the flow specifications to see if its capacity (in terms of bandwidth, buffer size, CPU speed, etc.) and its previous commitments to other flows can handle the new flow.

Internetworking

The basic functionality of network layer is to provide end – to – end communication capability on networks

It is the lowest layer that deals with end – to – end transmission. The network layer protocols are concerned with the exchange of packets of information between transport layer entities.

A packet is a group of bits that includes data bits plus source and destination address the service provided by the network layer to the transport is called network service.

The network layer functions are carried out by a layer adding a header to every network service data unit (NSDU) forming network protocol data unit (NPDU).

1. It keeps track which MAC, the unique number that each network card has address to send i.e. divide which system receiver the information.
2. It makes routing of data through network source to destination.
3. Virtual circuits are established in this layer
4. It translates logical network address into physical machine address.

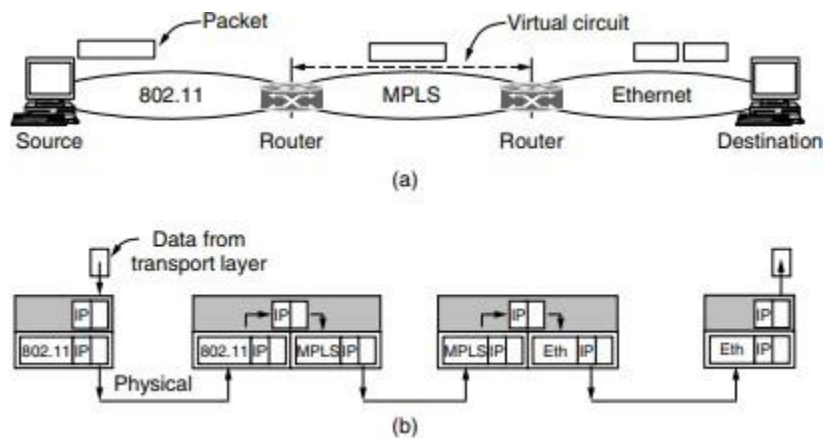
i. How Networks Differ:

Networks can differ in many ways. Some of the differences, such as different modulation techniques or frame formats, are internal to the physical and data link layers. These differences will not concern us here. Instead, in Fig. we list some of the differences that can be exposed to the network layer.

Item	Some Possibilities
Service offered	Connectionless versus connection oriented
Addressing	Different sizes, flat or hierarchical
Broadcasting	Present or absent (also multicast)
Packet size	Every network has its own maximum
Ordering	Ordered and unordered delivery
Quality of service	Present or absent; many different kinds
Reliability	Different levels of loss
Security	Privacy rules, encryption, etc.
Parameters	Different timeouts, flow specifications, etc.
Accounting	By connect time, packet, byte, or not at all

ii. How Networks are Interconnected:

We have discussed several different devices that connect networks, including repeaters, hubs, switches, bridges, routers, and gateways. Repeaters and hubs just move bits from one wire to another. They are mostly analog devices and do not understand anything about higher layer protocols. Bridges and switches operate at the link layer.



iii. **Tunnelling:** it is a strategy used when two computers want to communicate with each other and packet must pass through a region. In order to reach packet to the region it must have IP address.

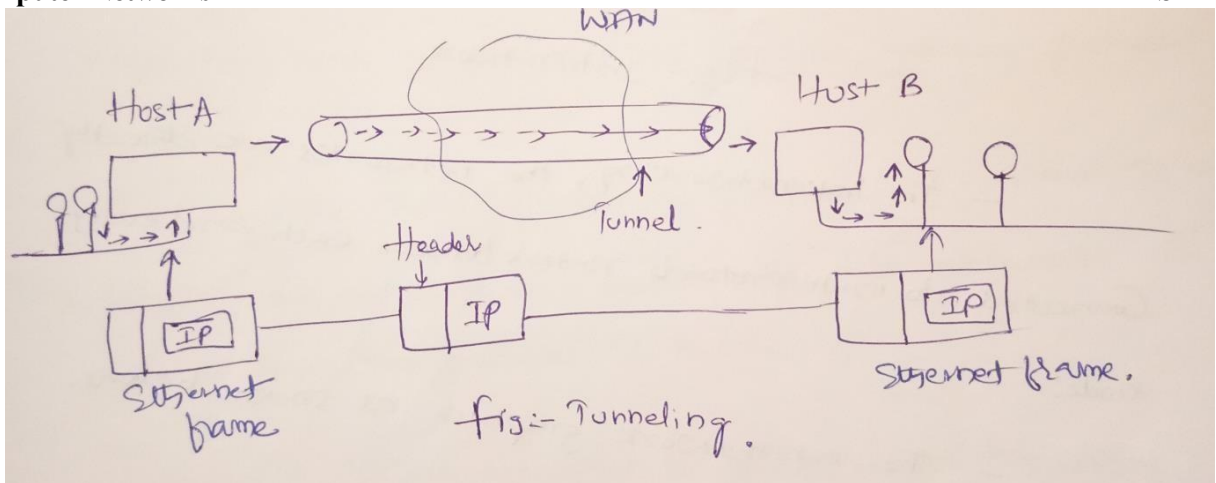
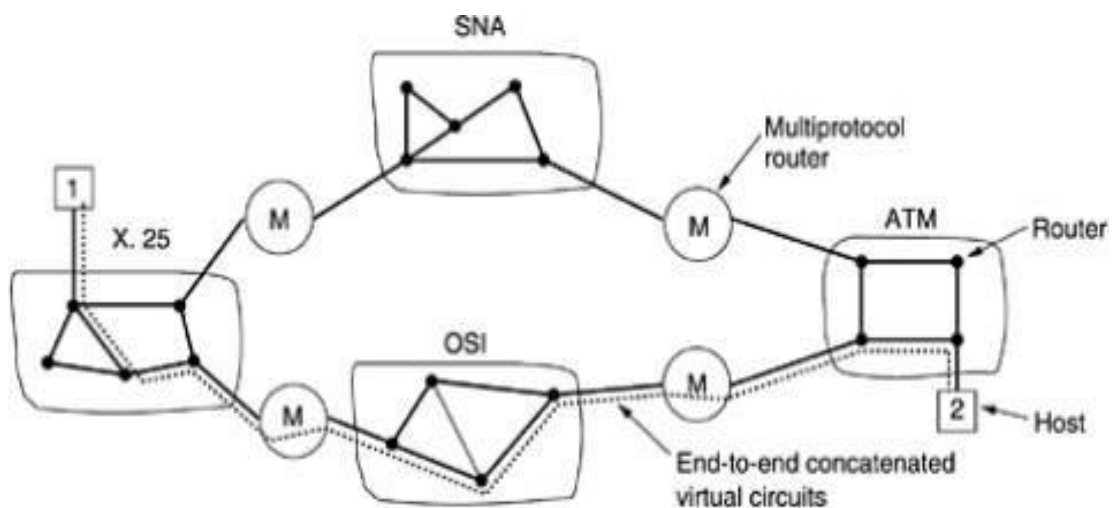
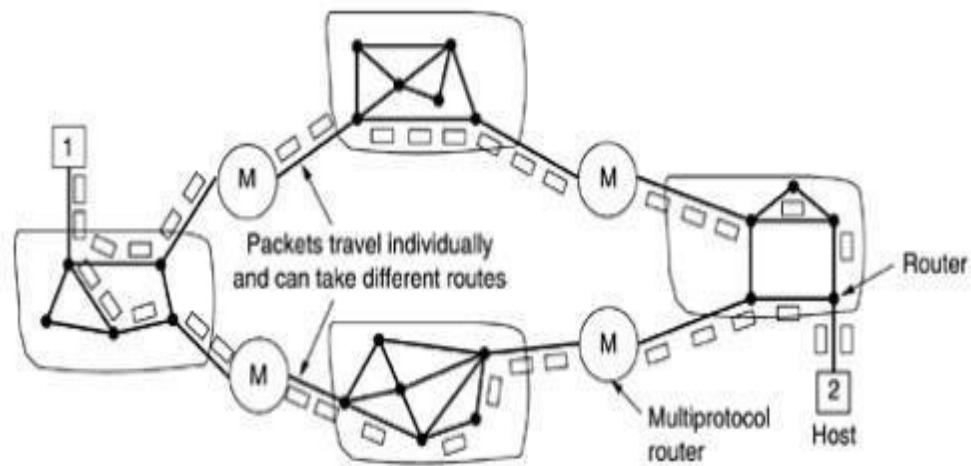
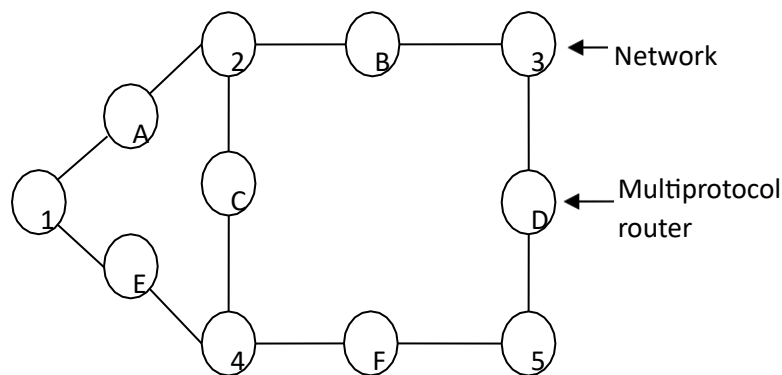


Figure:- Tunneling

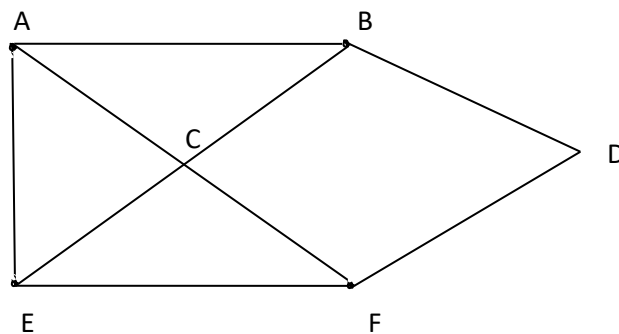
To send IP packet from host a to host B the packet contains IP address of host B. then the packet is transmitted from host A. When packet reaches to the router R, it removes IP packet. The router R1 insert a pay load field of the WAN network layer packet. After inserting pay load field it send to router R2. When router R2 gets the packet, it removes IP packet and send it to host B inside the header frame.

iv. Concatenated Virtual Circuit Internetworking



v. **Connection-less Interworking**vi. **Internetwork routing:****Figure: internetwork**

In internetworking the networks are directly connected to multi protocol routers between each and every node. The inter network graph is as shown below

**Figure: graph of internet work**

From an above figure where every router directly access every other router connected to any network which it is connected.

Router B cans directly access A and C via network 2 and also d via network 3.

While, constructing the graph of internetwork the distance vector routing algorithm is applied to the set of multiple protocols. Which may gives two level routing algorithm.

The first level is interior gateway protocol algorithm and second one is exterior gate way protocol algorithm.

The interior gate way protocol algorithm is used within each network and between the two networks an exterior gateway protocol is used.

The internet packets start out on its LAN addresses to the local multiprotocol router.

After reaching at router, the network layer code divides which multiprotocol router to forward the packet to using its own routing tables.

In a LAN when no. of users or stations are more and if the stations are located at a longer distances forming bigger LAN is not always possible due to following limitation.

1. Constraint of the maximum no. of stations which can be connected on a single LAN
2. Constraint of maximum physical size of a LAN
3. Network security
4. Investment already done and further to be done.

If LANs which are geographically separated or if they are needed to be connected or communicated, again these LAN's may not be of the same type i.e. one of these may be CSMA/CD LAN and other may be FDDI LAN.

The LANs and switched data sub networks are interconnected to allow the users to communicate across several sub networks. This extended network so formed is called internetwork.

Sub networks inter connected may differ in technologies topologies protocols and their services.

The internet work devices such as bridge, router are used to connected the two or more LANS or subnets.

Importance of internetworking:

1. It provides link between networks.

2. Provides routing and delivery of data between process on different networks.
3. Provides an accounting service that keeps track of the various networks and routers and that maintains status information.
4. Internetworking facility accommodate a number of different maximum packet size, difference access mechanism, error recovery, status reporting technique etc.

BGP Border Gateway Protocol (BGP):

BGP is an inter-domain routing protocol using path vector routing. It first appeared in 1989 and has gone through four versions. Types of Autonomous Systems As we said before, the Internet is divided into hierarchical domains called autonomous systems. For example, a large corporation that manages its own network and has full control over it is an autonomous system. A local ISP that provides services to local customers is an autonomous system.

We can divide autonomous systems into three categories: stub, multi-homed, and transit.

o **Stub AS:** A stub AS has only one connection to another AS. The inter-domain data traffic in a stub AS can be either created or terminated in the AS. The hosts in the AS can send data traffic to other ASs. The hosts in the AS can receive data coming from hosts in other ASs. Data traffic, however, cannot pass through a stub AS. A stub AS is either a source or a sink. A good example of a stub AS is a small corporation or a small local ISP.

o **Multi-homed AS:** A multi-homed AS has more than one connection to other ASs, but it is still only a source or sink for data traffic. It can receive data traffic from more than one AS. It can send data traffic to more than one AS, but there is no transient traffic. It does not allow data coming from one AS and going to another AS to pass through. A good example of a multi-homed AS is a large corporation that is connected to more than one regional or national AS that does not allow transient traffic.

o **Transit AS:** A transit AS is a multi-homed AS that also allows transient traffic. Good examples of transit ASs are national and international ISPs (Internet backbones).

Attributes are divided into two broad categories: **well known and optional**. A well-known attribute is one that every BGP router must recognize. An optional attribute is one that needs not be recognized by every router.

Well-known attributes are themselves divided into two categories: mandatory and discretionary. A well-known mandatory attribute is one that must appear in the description of a route. A well-known discretionary attribute is one that must be recognized by each router, but is not required to be included in every update message. One well-known mandatory attribute is ORIGIN. This defines the source of the routing information (RIP, OSPF, and so on). Another well-known mandatory attribute is AS_PATH. This defines the list of autonomous systems through which the destination can be reached. Still another well-known mandatory attribute is NEXT-HOP, which defines the next router to which the data packet should be sent.

The optional attributes can also be subdivided into two categories: transitive and non-transitive. An optional transitive attribute is one that must be passed to the next router by the router that has not implemented this attribute. An optional non-transitive attribute is one that must be discarded if the receiving router has not implemented it.

BGP Sessions: The exchange of routing information between two routers using BGP takes place in a session. A session is a connection that is established between two BGP routers only for the sake of exchanging routing information. To create a reliable environment, BGP uses the services of TCP. When a TCP connection is created for BGP, it can last for a long time, until something unusual happens. For this reason, BGP sessions are sometimes referred to as semi-pennant connections.

External and Internal BGP If we want to be precise, BGP can have two types of sessions: external BGP (E-BGP) and internal BGP (I-BGP) sessions. The E-BGP session is used to exchange information between two speaker nodes belonging to two different autonomous systems. The I-BGP session, on the other hand, is used to exchange routing information between two routers inside an autonomous system. Figure shows the idea.

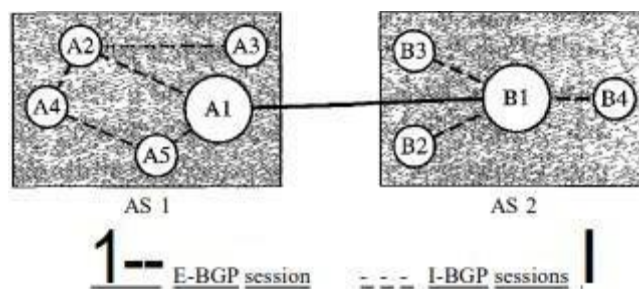


Fig. Internal and External BGP sessions

The session established between AS1 and AS2 is an E-BOP session. The two speaker routers exchange information they know about networks in the Internet. However, these two routers need to collect information from other routers in the autonomous systems. This is done using I-BOP sessions.

vii. Fragmentation:

Each network imposes some maximum size on its packets. These limits have various causes, among them:

1. Hardware (e.g., the size of an Ethernet frame).
2. Operating system (e.g., all buffers are 512 bytes).
3. Protocols (e.g., the number of bits in the packet length field).
4. Compliance with some (inter)national standard.
5. Desire to reduce error-induced retransmissions to some level.
6. Desire to prevent one packet from occupying the channel too long.

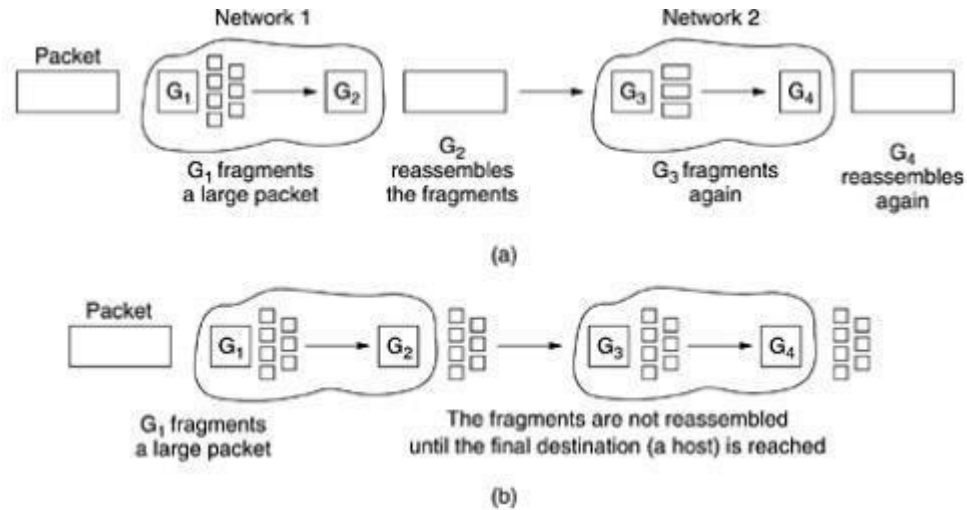
The result of all these factors is that the network designers are not free to choose any maximum packet size they wish. Maximum payloads range from 48 bytes (ATM cells) to 65,515 bytes (IP packets), although the payload size in higher layers is often larger.

An obvious problem appears when a large packet wants to travel through a network whose maximum packet size is too small. One solution is to make sure the problem does not occur in the first place. In other words, the internet should use a routing algorithm that avoids sending packets through networks that cannot handle them. However, this solution is no solution at all. What happens if the original source packet is too large to be handled by the destination network? The routing algorithm can hardly bypass the destination.

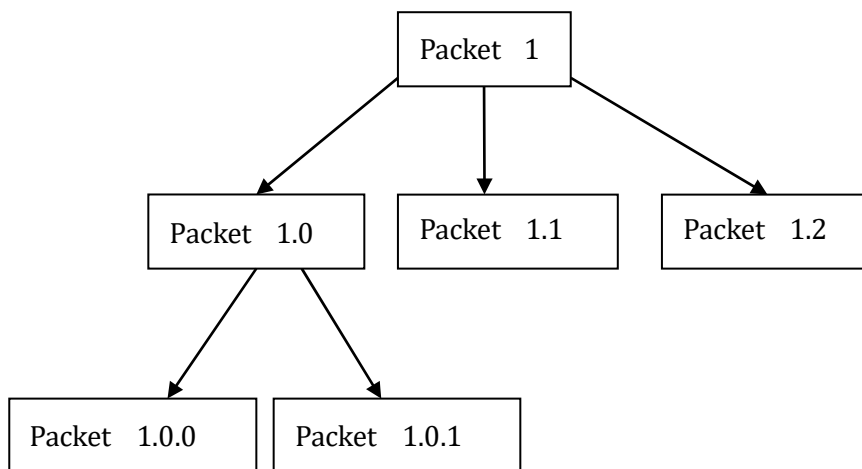
Basically, the only solution to the problem is to allow gateways to break up packets into fragments, sending each fragment as a separate internet packet. However, as every parent of a small child knows, converting a large object into small fragments is considerably easier than the reverse process. (Physicists have even given this effect a name: the second law of thermodynamics.) Packet-switching networks, too, have trouble putting the fragments back together again.

Two opposing strategies exist for recombining the fragments back into the original packet. The first strategy is to make fragmentation caused by a "small-packet" network transparent to any subsequent networks through which the packet must pass on its way to the ultimate destination. This option is shown in Fig. 5-50(a). In this approach, the small-packet network has gateways (most likely, specialized routers) that interface to other networks. When an oversized packet arrives at a gateway, the gateway breaks it up into fragments. Each fragment is addressed to the same exit gateway, where the pieces are recombined. In this way passage through the small-packet network has been made transparent. Subsequent networks are not even aware that fragmentation has occurred. ATM networks, for example, have special hardware to provide

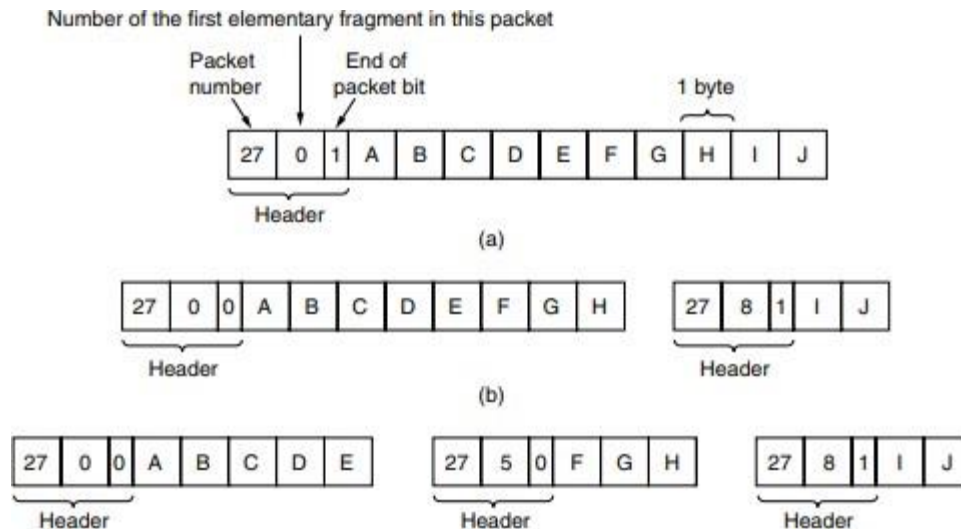
transparent fragmentation of packets into cells and then reassembly of cells into packets. In the ATM world, fragmentation is called segmentation; the concept is the same, but some of the details are different.



Numbering Individually Fragments: It can be done in two ways: Using Tree Structure:



Defining Elementary Fragment sizes: Using this type each packet will be fragmented equally except the last one.



The network layer in Internet:

IP address:

The IP address corresponds to the network layer in the OSI reference model and provides a connection less. Best effort delivery services to the transport layer. An internet protocol has a fixed length of 32 bits.

The IP address is unique; the two devices on the internet can never have the same address at the same time. The IP address structure was originally defined to have a network id and host id. The network id identifies the network the host is connected to the host id identifies the network connection to the host rather than the actual host.

An address space is the total no. of addresses used by the protocol. If protocol used N bits to define an address the address if protocol uses $N -$ bits to define an address, the address space is 2^N .

IP address is usually written in dotted decimal notation so that they can be communicated conveniently by people.

The IP address is broken into four bytes with each byte being represented by a decimal number and separated by a dot.

Example: 192.168.1.2 is a dotted decimal notation. The address 192.168.1.2 in a binary notation is given by

27	26	25	24	23	22	21	20
128	64	32	16	8	4	2	1

An IP is a numeric identifier assigned to each machine on IP network

1 1 0 0 0 0 0 0	1 0 1 0 1 0 0 0	0 0 0 0 0 0 0 1	0 0 0 0 0 0 1 0
192	168	1	2

An IP address is a software address not a hardware address which is loaded in the machine NIC

An IP address is made up of 32 bits of information

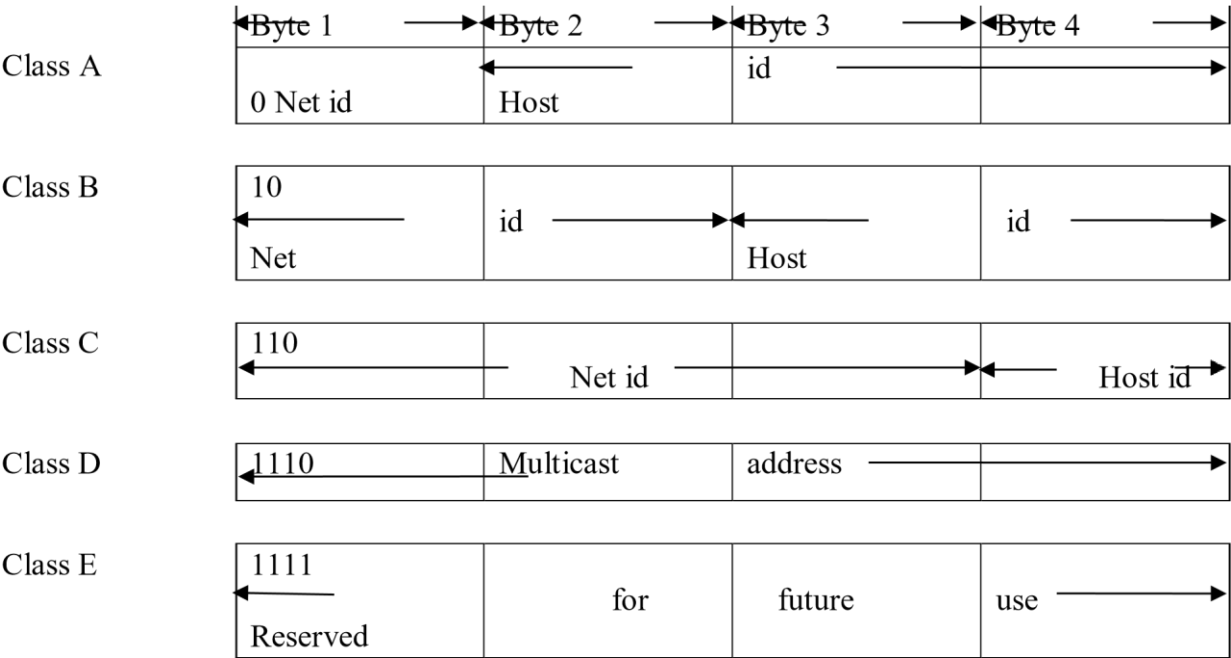
These bits are divided into four parts containing 8 – bits each.

Every machine on the same network shares that network address as part of its IP address.

The IP address 192.168.1.2 the bytes 192.168 is the network address and 1.2 is the node address.

The node address is uniquely identifies each machine on network.

Internet classes (IP address):



Ranges:

	From		To
Class A	0.0.0.0.		127.255.255.255
Class B	128.0.0.0		191.255.255.255
Class C	192.0.0.0		223.255.255.255
Class D	224.0.0.0		239.255.255.255
Class E	240.0.0.0		255.255.255.255

The IP address structure is divided into five classes as shown in above Class A, B, C, D & E.

In class A address are used for large sized organization with a large no. of attached hosts on routers.

In class B address were designed for mid-sized organizations with tens of thousands of attached hosts or routers.

In class C addresses were designed for small sized organizations (Home, offices and corporate companies).

Class D addresses are used for multicast services that allow a host to send information to a group of hosts simultaneously. Class E reserved for the future use.

In class full addressing each class is divided into a fixed no. of blocks with each block having a fixed size.

Class	No. of Blocks	Blocks size
A	128	1677216
B	16384	65536
C	2097152	256
D	1	268435456
E	1	268435456

Classless Addressing

In classless addressing variable length blocks are assigned they belong to no class. In this entire address space is divided into blocks of different sizes

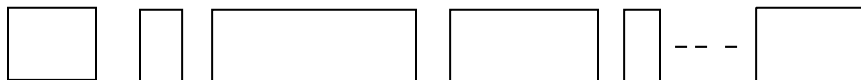


Figure: class less addressing

In classless addressing, when an entity, small or large needs to be connected to the internet it is granted a block of address.

The size of the block varies based on the nature and size of the entity.

In IPV4 addressing a block of address can be defined as 192.168.1.2/8 in which 192.168.1.2 defines one of the address and /8 is define the mask on no. of bits.

The IPV4 is the delivery mechanism used by TCP / IP protocols. IPV4 is an unreliable connectionless datagram protocol

IPV4 is also a connection less protocol for packet switching network that uses the datagram approach.

CIDR (Class Inter Domain Routing):

The addresses that are divided into A, B, C, & D classes turned out be inflexible. IP is rapidly becoming a victim of its own popularity, is running out of addresses.

In 1993 the classful address space restriction was lifted. An arbitrary prefix length to indicate the network number known as classless inter-domain routing (CIDR) adopted in place of classful scheme. Using a CIDR notation, a prefix 205.100.0.0 of length 22 is written as 205.100.0.0/22.

The corresponding prefix range from 205.100.0.0 through 205.100.3.0 (the /22 notation indicates that the network mask is 22 – bits. CIDR routes packets according to the higher order bits of the IP address. The entries in a CIDR routing table contains a 32 – bit address and a 32 – bit mask.

CIDR uses a technique called super netting so that a single routing entry covers a block of classful addresses.

For Example: address of class C i.e. 205 .100.0.0, 205.100.1.0, 205.100.2.0,205.100.3.0, the CIDR allow a single entry 205.100.16.0/22

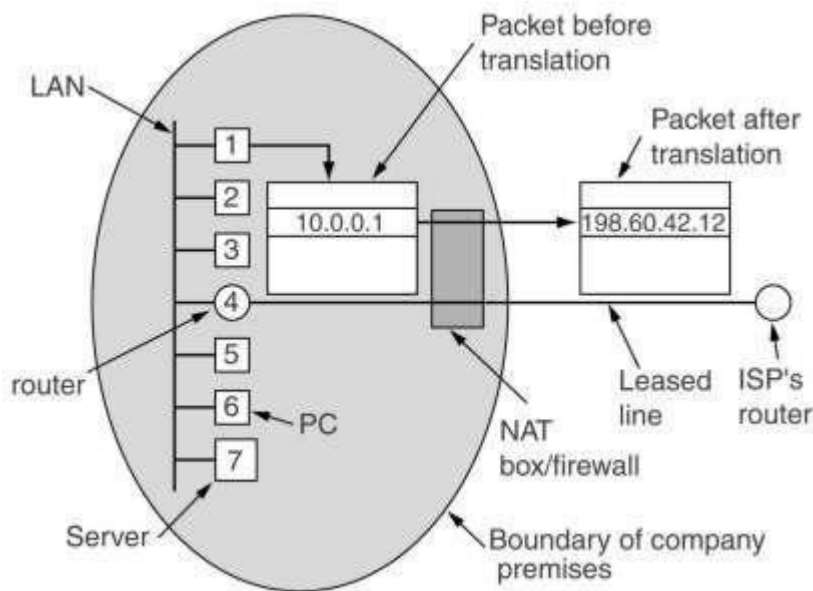
The use of the variable length prefixed requires that the routing tables be searched to find the longest prefix match.

NAT (Network Address Translation)

IP addresses are scarce. An ISP might have a /16 (formally class B) address, giving 64K host numbers. If it has more customers than that, it might give each customer with a dial-up connection an dynamically assigned IP number when it logs in. Normally not all customers are logged on at the same time.

Computer Networks

SIET



Another way out is using NAT. Within a company internal IP addresses of the form 10.x.y.z are used. When communicating with the outside world, all internal IP numbers are changed to an external number. The problem occurs when a packet comes back (e.g. from a WEB server). How does the NAT know which internal IP number to use? Many internal computers might be communicating with the same WEB server.

The dirty solution is that most IP packets carry a TCP or UDP payload. Their headers contain a field, a 16 bit **source port**, which indicates which process on the sending machine has send the TCP or UDP payload. The receiving machine uses this port number, now placed in the **destination port** to send information back. On sending the NAT replaces the source port with an index to a table, containing the real internal IP and source port. On receiving it looks up that information and changes the incoming packet accordingly.

The method as important disadvantages. It violates the whole hierarchical architectural model. It makes the IP network in fact connection-oriented as it maintains information on each connection passing through it. A crash of the NAT box terminates every TCP connection. Further there are some protocols, like FTP and the H.3232 Internet telephony protocol, which send IP numbers (and port numbers) in data, to be used by the other side. These protocols fail unless certain precautions are taken or patches are installed in the NAT software.

NAT is useful as an intermediate solution until version 6 of IP, with a much larger IP bursty space, is in widely use.

Computer networks logical addressing: In this article, we are going to learn about the IPV4 addresses which are part of logical addressing.

Usually, computers communicate through the Internet. The packet(data) transmitted by the sender computer may pass through several LANs or WANs before reaching the destination computer. For this level of communication, we need a global addressing scheme what we call logical addressing. An IP address is used globally to refer to the logical address in the network layer of the TCP/IP protocol.

The Internet addresses are 32 bits in length; this gives us a maximum of 2^{32} addresses. These addresses are referred to as IPv4 (IP version 4) addresses or popularly as IP addresses.

IPv4 addresses

An IPv4 address is a 32-bit address that uniquely and universally defines the connection of a device (for example, a computer or a router) to the Internet. They are unique so that each address defines only one connection to the Internet. Two devices on the Internet can never have the same IPV4 address at the same time.

On the other hand, if a device operating at the network layer has m connections to the Internet, it needs to have m addresses, for example, a router.

The IPv4 addresses are universal in the sense that the addressing system must be accepted by any host that wants to be connected to the Internet. That means global addressing. Address Space:

IPv4 has a certain address space. An address space is the total number of addresses used by the protocol. If a protocol uses N bits to define an address, the address space is 2^N . IPv4 uses 32-bit address format, which means that the address space is 2^{32} or 4,294,967,296. The Notations are show an IPv4 address:

1) Binary Notation

In binary notation, the IPv4 address is displayed as 32 bits. Each octet is often referred to as a byte. So it is common to hear an IPv4 address referred to a 4-byte adder

ss. The following is an example of an IPv4 address in binary notation: 01110111 10010101 00000001 00000011.

2) Dotted-Decimal Notation

IPV4 addresses are usually written in decimal form with a decimal point (dot) separating the bytes since it's more compatible. The following is an example: 119.149.1.3 (above one and this one is same just different notation) : Each number in dotted-decimal notation is a value ranging from 0 to 255.

IPV4 Header Format:

Packets in the IPVH layer are called datagram's.

A datagram is a variable length packet consisting of two parts. Header and data.

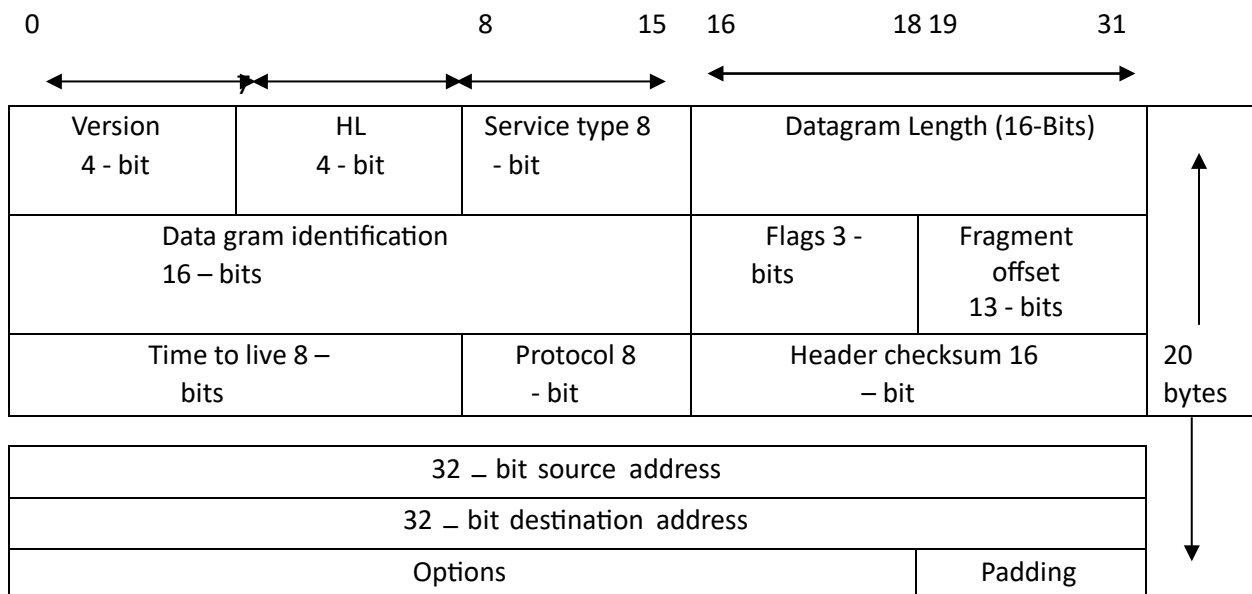
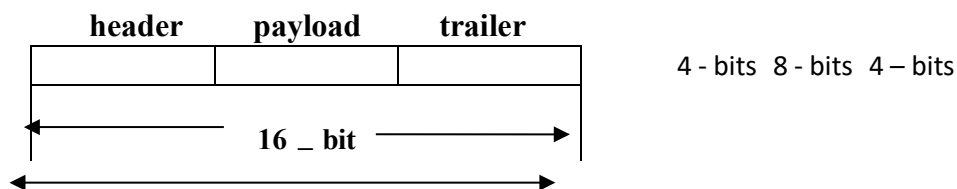


Figure IPV4 header format



Datagram

1. **Version:** it is the field contains the IP protocol version the current version is an experimental version. IPVH (or) IPVH.S
2. **IHL (IP Header Length):** it is the length of the ip header in multiples of 32 – bits without the data field. The maximum value of header is (60 bytes) and minimum value of (20 bytes)
3. **Service type:** it is a differentiated service the IP monitors the quality of the service of datagram.
4. **Total length:** it is specifies the total length of the data gram, header and data, in octets.
5. **Datagram identification:**

Which fragments the packets a unique numbers assigned by the sender.

6. **Flogs: it contains 3 flags:**

1. The flag is zero – it is reserved for zero.
2. The 2nd flag is DF (Do not fragment flag) ‘0’ means allow fragmentation and set to ‘1’ means no fragmentation allowed
3. The 3rd last flag is MF (more fragment flag). ‘0’ means that is the last fragment. And set to ‘1’ means more fragmentation of packets.

7. **Fragment offset:**

It is used to reassemble the full datagram. The value in this field contains the no. of zero.

8. **Time – to – live (TTL):**

It specifies the time (in Secs) the datagram allowed to travel. It uses as hop counter to detect routing loops.

9. **Protocol:** the IPVH protocol is in the TCP segment. It indicates that the protocol high level to which IP should the data in this datagram. (ICMP =1, TCP = 6, UDP = 17, OSPF=89)

10. **Header Checksum:** it is checksum for information contained in the header. If the checksum does not match the contents, the datagram is discarded.

11. **Source and Destination Address:**

It is a 32 – bit source and destination address in which the datagram are to be forwarded successfully

12. **Options:** it is a variable length field used for control (or) debugging and measurement.

13. **Padding:** it is used to ensure that the IP header ends on a 32 – bit boundary.

IPV6:

IPV4 provides the host to host communication between system in the internet. In early 1990 the IETF began to work on the successor of IPV4 that would solve the address exhaustion problem and other scalability.

IPV6 addresses are 128 – bits in length. Addresses are assigned to individual interface on nodes, not to the node themselves.

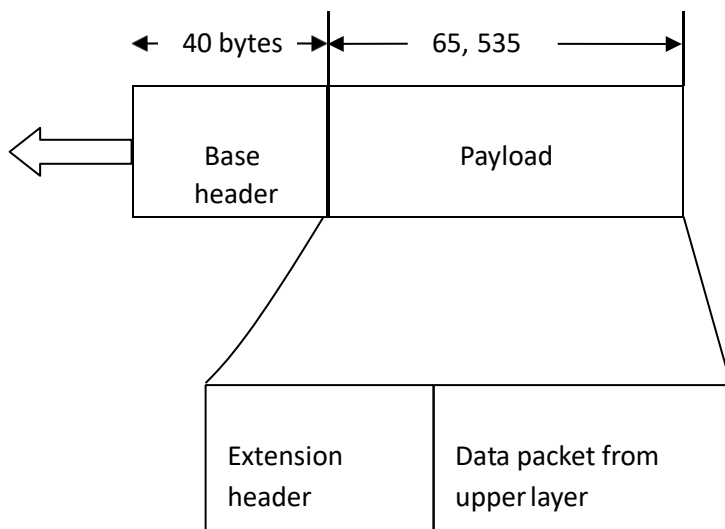
A single interface may have multiple unique unicast address. The first field of any IPV6 address is the variable length prefix format.

IPV6 addresses: the new notation has been written in 16 – bytes address format. It can be eight groups of four hexadecimal digits with colons between the group 8000: 0000: 0000: 0000: 0123 : 4567 : 89AB; CDEF

The leading zeros within a group can be omitted so 0123 can be written as 123. One or more groups of 16 zero bits can be replaced by a pair of colons.

The new address is given as 8000: 0123: 4567: 89AB: CDEF.

IPV 6 Packet Format: The IPV6 each packet is composed of base header followed by the pay load.



IPV6 Header Format:

Version 4 - bits	Traffic class 4 - bits	Flow label 24 - bits		
Pay load length 16 – bits			Next header 8 - bits	Hop limit 8 - bits
128 - bit source address				
128 – bits destination address				

Figure IPV6 header format

1. **Version:** this 4 – bit field defines the version number of IP. The value is 6 for IPV6
2. **Traffic Class:** In this field the first MSB of 6 – bit provides the quality of service and remaining 2 – bits which can use for explicit congestion control (ECN)
3. **Flow Label:** it is 24 – bits field that is designed to provide sequential flow of data for particularly.
4. **Pay Load Length:** the 16 bits pay load length field defines the lengths of the ip data gram excluding the base header. The pay load length is starts from 16 – bit and its extend up to 65, 535 bytes.

Next Header: it is 8 – bit field defining the header that follows the base header in the datagram.

It indicates that header (or) extension if the extension header is not present then it indicates the upper layer.

Hop Limit: It is the 8 – bit field, used to stop packet to loop in the network This is same as that of TTL (Time to Live) in IPV4

The value of hop limit field is decremented by 1 as it passes a link (router/hop) when field reaches 0 the packet is discarded.

Source Address: this field is 128 – bits indicate that address of originator of the packet.

Destination Address: this field provides the address of intended recipient of the packet.

Extension Headers: in IPV 6 the fixed header contains only that much information which is necessary, avoiding those information which is either not required or is rarely used.

All such information is put between the fixed header and the upper layer header in the form of extension headers.

The Following Extension Header:

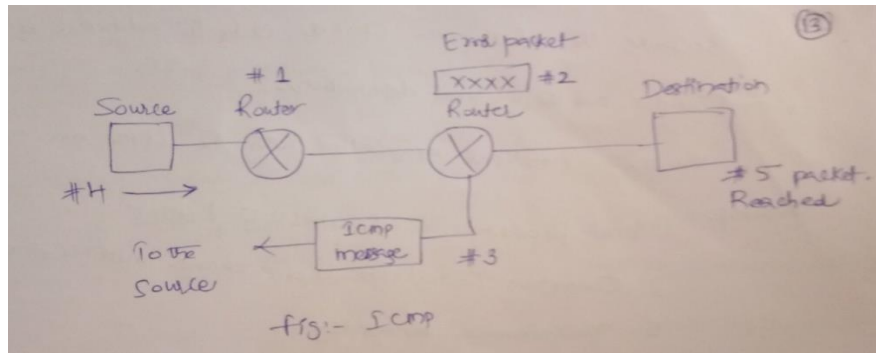
Extension header	Next header value	Description
Hop – by – hop option header	0	Read by all devices to pass info
Routing header	43	If combines the methods to support making routing decisions.
Fragment header	44	Contains parameters of datagram fragmentation.
Destination options header	60	Used when heads by destinations devices
Authentication header	51	Information regarding authentication.
Encapsulating security payload header	50	It provides confidentiality and ensure integrity of data.

ICMP (Internet Control Message Protocol):

ICMP is companion protocol to IP. These two are implemented together. It is to sit on the top of the IP (IP protocol =1).

It allows routers to send error or control messages to other routers or hosts; it provides communication between the internet protocol software on the machine and the internet protocol software another.

It is an error reporting and testing mechanism. It provides a way for router that encounters an error report the error to the original source.



ICMP Message Format:

8 - bits	8 - bits	16 - bits
Type	Code	Checksum

When over packets sending from source, the one of the routers which may treated as error packet and which are discarded by the router and ICMP sends an error report to the original source. The source itself modify the error packet when forwarded to the destinations.

The ICMP sends message by encapsulating them in IP packets and setting the header protocol by 1.

The ICMP can send only to the source, not to an intermediate router. Because the datagram carries only the addresses of the original sender and the final destination.

The main functions associated with the ICMP are as follows.

1. Error reporting
2. Reach ability testing
3. Congestion control
4. Route change notification
5. Performance measuring The

ICMP messages are given by

Name	Type / Code	Usage
Destination unreachable (node, net, and host)	3/0 (or) 1	Lack of connectivity

Destination unreachable (fragments)	3/4	Path discovery MTU
Time exceeds	11/0	Trace router
Echo request (or) reply	8 or 0	ping

Destination unreachable means packet is not delivered to the destination router cannot find the destination.

Time exceeded is used when time to live field hits to zero. Life time of data gram expires. Echo replay means the device in the network is alive.

ARP (address resolution protocol):

The address resolution protocol (ARP) is used to associate a logical address with physical address.

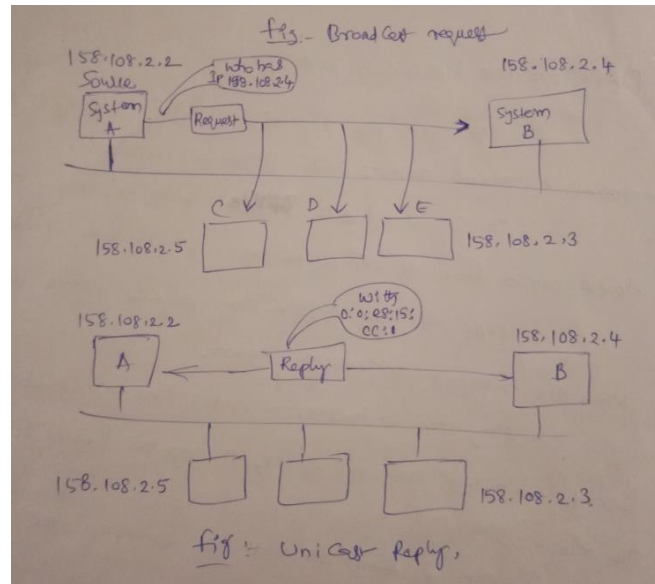
On typical physical networks such as LAN, each device on a link is identified by a physical or station address when its internet address is known.

Any time a host or a router has an IP datagram to send to another host or router, it has the logical IP address of the receiver.

An IP datagram must be encapsulated in a frame to be able to pass through the physical network. This means that the sender needs the physical address of the receiver.

A mapping corresponds a logical address to a physical address. The mapping can be done dynamically which means that the sender asks receiver to announce its physical address when needed.

It always sends a ARP query packet. The packet includes the physical address and IP address of the sender and the IP address of the receiver.

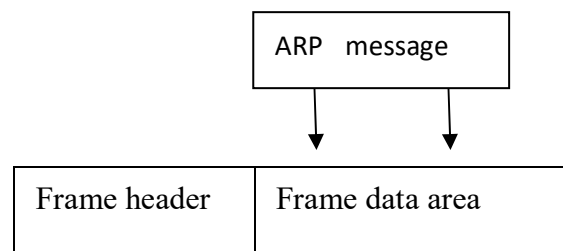


Consider a example computer A and computer B share a share physical network. Each computer has an assigned IP address and physical address.

Address mapping must be performed at each step along the path from the original source to the destination. The sender must map the intermediate routers internet address to a physical address. From on above broad cast ARP host A broadcast an ARP request containing system B IP address to al computer on the network.

Figure (2) shows that host B responds with an ARP reply that contains the pair (IP at B and physical address at B) whenever a computer receives an ARP reply it saves the sender's IP address and its corresponding address in its cache. When transmitting a packet, a computer always looks in its cache for a binding before sending an ARP request.

If a computer finds the desired binding in its ARP cache, it need not broadcast on the network. When ARP message travel from one computer to other they must be carried in physical frame.



The above figure shows that the ARP message is carried in the data portion of a frame.

ARP Packet format:

Hardware Type: this is the 16 – bit field defining the type of network on which ARP is running. Ethernet value is 1

Protocol Type:

This is the 16 – bit field defining the protocol the value of this field for the IPV4 protocol is 0800H.

32 - bits			
Hardware type		Protocol type	
Hard length	Protocol length	Operation	
		1. request	2. reply
Sender protocol address			
Target hardware address			
Target protocol address			

Hardware Length: this is an 8 – bit field defining the length of the physical address in bytes Ethernet value is 6.

Protocol Length: This is an 8 – bit field defining the length of logical address in bytes for the IPV 4 protocol the value is 4.

Operation: it is defining the type of packet

1. APR Request
2. APR reply

Sender Hardware Address: this is variable length field defending the physical address of the sender (6 – byte long)

Sender Protocol Address: this is also a variable length field defining the logical address of the sender (4 – bytes long)

Target Hardware Address: this is a variable length field defining the physical address of the target. (6 – bytes long)

Note: for ARP request message, this field is all 0's because the sender doesn't know the physical address of the target.

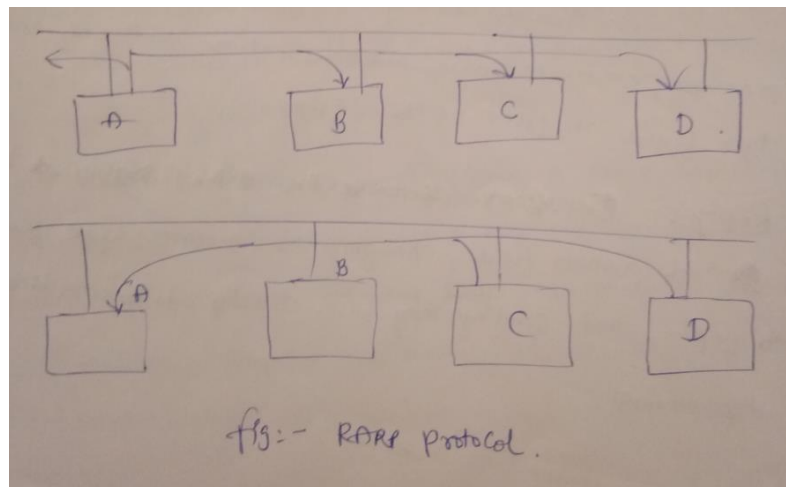
Target Protocol Address: It is also a variable length field defining the logical address of the target. (4 – bytes long)

RARP (Reverse Address Resolution Protocol):

In some of the instances, a host may know its MAC address but not its IP address.

For example, when a diskless computer such as an a terminal is being bootstrapped, it can send the MAC address from its Ethernet card.

Its IP address is usually kept separately in a disk at the server. The problem of getting an IP address from MAC address can be handled by RARP, which works on the fashion of similar to ARP



The sender broadcast a RARP request that specifies itself as both. The sender and target machine and supplied its physical address in the target hardware address filled.

All the machines on the network receive the request, but only those authorized to supply the RARP service process the request and send a reply, such machines are known as RARP server. If RARP succeeded means, the network must contain at least one RARP server.

Request: like ARP messages, a RARP message is sent from one machine to the another encapsulated in the data portion of a network frame.

A frame carrying a RARP request has the usable preamble, Ethernet source and destination addresses, and packet type field in front of the frame.

Reply: Server answers request by filling in the target protocol address field, changing the message type from request to reply, and sending reply back directly to the machine making the request.

DHCP (Dynamic Host Control Protocol):

DHCP provides static and dynamic address allocation that can be manual or automatic.

DHCP does not require an administrator to add an entry for each computer to the data base that a server uses.

DHCP provide mechanism that allows a computer to join a new network and obtain IP address without manual invention.

The DHCP works like a plug & play networking

DHCP is in wide use because it provides a mechanism for assigning temporary IP network address to hosts. This capability is used extensively by internet service providers to maximum the usage of their limited IP address space.

DHCP has a pool of available IP address when a DHCP client request a temporary IP address the DHCP server goes to the pool of available IP addresses (unused IP address) and assigns an IP address for a negotiable period of time. An administrator can configure a DHCP server to have two types of address. Permanent addresses to be allocated on demand. When a computer boots and send a request to DHCP the DHCP server consults its database to find configuration information.

DHCP uses a same technique as Bootp: each computer waits a random time before transmitting or retransmitting a request

When a host wishes to obtain an IP address the host broadcasts a DHCP discover message in its physical network.

The server in the network may respond with a DHCP offer message that provides an IP address and other configure information.

When a computer discover a DHCP server the computer saved the servers address in a cache on permanent storage once it obtains an IP address, the computer saves the IP address in cache.

